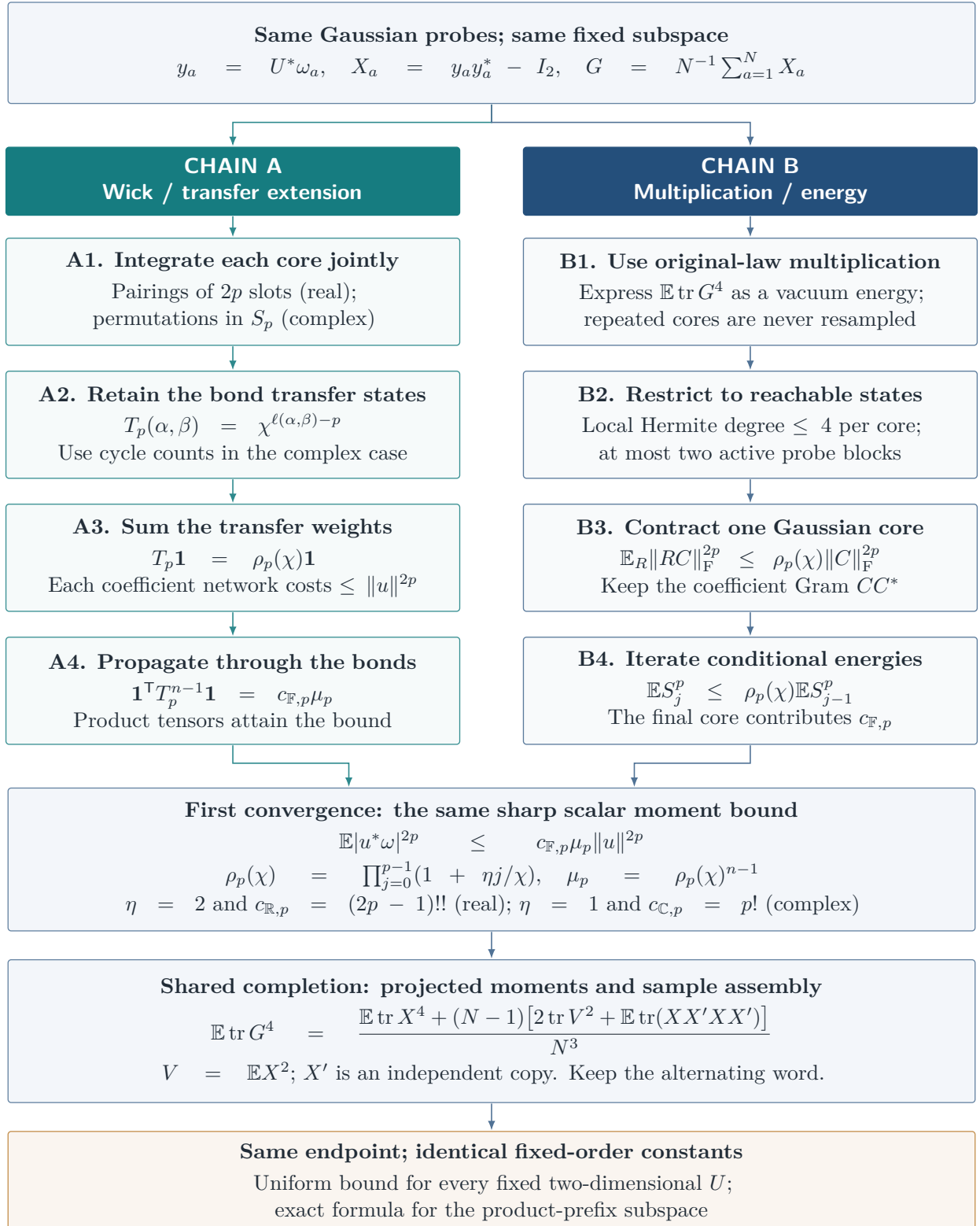


Gaussian rMPS moments

Where the proof chains split and meet

Read downward. Teal and blue show the two routes; full-width boxes show the shared input, convergence, and common completion.



Three-chain comparison

Two routes to one moment theorem; a hybrid reconstruction application.

Chains A and B have the *same* rMPS endpoint and constants. Chain C combines Wick contractions with seeded Hermite multiplication for a *different* quantity: an inverse Gram coupled to the residual from the same sketch.

Comparison	Chain A (§2) Wick / transfer	Chain B (§3) Multiplication / energy	Chain C (§§8–13) Mixed Wick–Hermite
Quantity	Centered Gram moment $\mathbb{E} \operatorname{tr} G^4$ for independent rMPS probes.	The same $\mathbb{E} \operatorname{tr} G^4$ for the same probe law.	Reconstruction penalty $F = c^\top \Gamma^{-2} c$; $\Delta = \Gamma - I_r$.
Geometry and field	Real or complex cores; every fixed two-dimensional U . Exact formula on the product-prefix family.	Exactly the same model and subspace scope as A.	Real shared-factor Khatri–Rao sketch; specified orthogonal signal–residual family. Rank-two certificates and fixed-rank expansion.
Retained information	Core pairings or permutations, bond-loop counts, and coefficient-network contractions.	Reachable Hermite sectors and the coefficient Gram across each conditional core contraction.	The seed c , its live sample label, and signed mixed moments $M_j = \mathbb{E}[c^\top \Delta^j c]$ through $j = 4$.
Wick’s role	Exact joint core integration; transfer row sum $T_p \mathbf{1} = \rho_p(\chi) \mathbf{1}$.	Not needed to enumerate the core states; Gaussian rotation and convexity give the same factor $\rho_p(\chi)$.	Evaluates marked sample partitions and coefficient contractions without separating Δ from c .
Framework role	A standalone transfer proof; no graded cutoff is needed.	Original-law vacuum energy, exact reachable-state cutoff, one-core contraction, and energy assembly.	Seeded finite Hermite compression; even moments as energies and odd moments as cross inner products.
Completion and inverse control	Spherical averaging and ordered fourth-order sample assembly. No inverse is present.	The same shared completion as A; no inverse-tail input is needed.	Signed finite polynomial majorant on a good event. A separate inverse moment bound justifies the un restarted expectation.
Constants and gain	Scalar eighth-moment factors $105\mu_4$ (real), $24\mu_4$ (complex); identical final constants to B.	Same scalar, projected, and centered-Gram constants as A; not a constants improvement over A.	At rank two, the true correction is $-30/m^2$, not the $+198/m^2$ produced by independent replacement.
Scope limit	Fixed fourth centered matrix moment; not a sharp growing-rank or logarithmic-confidence theorem.	Same limit as A. This example does not establish stronger bounds than Wick/transfer.	One specified geometry; not arbitrary-geometry universality or algorithmic optimality. Polynomial moments alone do not control inverses.

Quantitative endpoints (real case)

$$\text{A and B: } \mathbb{E} \operatorname{tr} G^4 \leq \frac{384\mu_4 + 3(N-1)(8\mu_2 - 2)^2}{N^3},$$

$$\text{C: } \mathbb{E} F = \frac{6}{m} - \frac{30}{m^2} + \frac{2598}{m^3} + O(m^{-7/2}).$$

N counts probes; m counts sketch rows. The second line is the rank-two, unit-residual expansion with separate inverse control; it is not a bound for arbitrary rMPS geometry.

1 Model, theorem, and comparison at a glance

1.1 The unchanged Gaussian rMPS model

Fix integers $n \geq 2$, $d \geq 2$, $\chi \geq 1$, and $N \geq 1$. The tensor length is n , the physical dimension is d , the bond dimension is χ , and the number of independent probes is N . The ambient dimension is d^n , not N . Write $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, with $z \sim N_{\mathbb{C}}(0, 1)$ meaning $z = (g + ih)/\sqrt{2}$ for independent real standard Gaussians.

Let the independent standard Gaussian cores have sizes

$$H_1 \in \mathbb{F}^{d \times \chi}, \quad H_j \in \mathbb{F}^{\chi \times d \times \chi} \quad (2 \leq j < n), \quad H_n \in \mathbb{F}^{\chi \times d}.$$

Define the random tensor by

$$\omega(i_1, \dots, i_n) = \chi^{-(n-1)/2} \sum_{\alpha_1, \dots, \alpha_{n-1}=1}^{\chi} H_1(i_1, \alpha_1) \left[\prod_{j=2}^{n-1} H_j(\alpha_{j-1}, i_j, \alpha_j) \right] H_n(\alpha_{n-1}, i_n). \quad (1.1)$$

The product over interior sites is empty when $n = 2$. This is the normalization of Definition 1.1 of Camaño–Epperly–Meyer–Tropp [1], with its deterministic scaling collected into a single factor. Gaussian tensor-train random projections already appear in Rakhshan–Rabusseau [2]; the model is not introduced in this note. Equivalently, multiply each of the first $n - 1$ standard cores by $\chi^{-1/2}$ and leave the last one unscaled. Both descriptions give the same random tensor.

Fix a deterministic isometry $U : \mathbb{F}^2 \rightarrow \mathbb{F}^{d^n}$, and let $\omega_1, \dots, \omega_N$ be independent copies of (1.1). Set

$$y_a = U^* \omega_a, \quad Y_a = y_a y_a^*, \quad X_a = Y_a - I_2, \quad G = \frac{1}{N} \sum_{a=1}^N X_a. \quad (1.2)$$

Every statement is for *each fixed* U , with constants independent of its coefficients. It is not a simultaneous statement over all subspaces chosen after observing the probes. Unmarked matrix norms are operator norms.

Direct contraction of two copies of the cores gives

$$\mathbb{E} \omega \omega^* = I_{d^n}, \quad \mathbb{E} Y_a = I_2, \quad \mathbb{E} X_a = 0. \quad (1.3)$$

Indeed, the physical and bond indices of the two copies must agree. The χ^{n-1} surviving bond paths cancel the squared normalization.

1.2 Moment factors and the two shared endpoints

For integers $p \geq 1$, define

$$\eta = \begin{cases} 2, & \mathbb{F} = \mathbb{R}, \\ 1, & \mathbb{F} = \mathbb{C}, \end{cases} \quad \rho_p(\chi) = \prod_{j=0}^{p-1} \left(1 + \frac{\eta j}{\chi} \right), \quad \mu_p = \rho_p(\chi)^{n-1}. \quad (1.4)$$

Set $\rho_0 = \mu_0 = 1$. In particular, $\mu_1 = 1$.

Theorem 1.1 (Uniform two-dimensional endpoint). *For every fixed two-dimensional isometry U , the real-core model satisfies*

$$\mathbb{E} \operatorname{tr} G^4 \leq \frac{384\mu_4 + 3(N-1)(8\mu_2 - 2)^2}{N^3}. \quad (1.5)$$

For circular complex Gaussian cores,

$$\mathbb{E} \operatorname{tr} G^4 \leq \frac{120\mu_4 + 3(N-1)(6\mu_2 - 2)^2}{N^3}. \quad (1.6)$$

Both proof chains give these same constants. No coefficient-sign condition, common Schmidt basis, or bound on the Schmidt ranks of U is required.

Define the product-prefix subspace

$$U_0 e_1 = e_1^{\otimes(n-1)} \otimes e_1, \quad U_0 e_2 = e_1^{\otimes(n-1)} \otimes e_2. \quad (1.7)$$

Its two coordinates share the first $n - 1$ cores; their dependence is not discarded.

Theorem 1.2 (Exact product-prefix endpoint). *For $U = U_0$, write $a = \mu_2$, $b = \mu_3$, and $c = \mu_4$. Then*

$$\mathbb{E} \operatorname{tr} G^4 = \frac{A_{\mathbb{F}} + (N - 1)B_{\mathbb{F}}}{N^3}, \quad (1.8)$$

where

$$A_{\mathbb{R}} = 384c - 192b + 48a - 6, \quad B_{\mathbb{R}} = 88a^2 - 48a + 6, \quad (1.9)$$

$$A_{\mathbb{C}} = 120c - 96b + 36a - 6, \quad B_{\mathbb{C}} = 48a^2 - 36a + 6. \quad (1.10)$$

The corresponding second-moment identities are

$$\mathbb{E} \operatorname{tr} G^2 = \begin{cases} (8a - 2)/N, & \mathbb{F} = \mathbb{R}, \\ (6a - 2)/N, & \mathbb{F} = \mathbb{C}. \end{cases} \quad (1.11)$$

1.3 What is actually being compared

Feature	Chain A: Wick / transfer	Chain B: multiplication / energy
Local description	Pairings or permutations for repeated Gaussian occurrences.	The original-law vacuum state, core degrees, and retained coefficient matrices.
Bond estimate	Exact transfer row sum $\rho_p(\chi)$.	Conditional Frobenius-energy factor $\rho_p(\chi)$.
Coefficient control	Contract the coefficient network and apply Frobenius Cauchy–Schwarz.	Form CC^* , then apply convexity to its eigenvalue weights.
Common output	Sharp scalar moments; spherical averaging; the exact ordered fourth-moment identity for independent sample blocks.	
Explicit constants	Identical in both endpoint theorems and all reported fourth-moment certificates.	
Limitation	The source paper’s fourth-moment surrogate cannot simply be used at eighth order.	The graded description is compatible with the proof, but its specialized machinery is not necessary for this fixed-order result.

1.4 Reading guide and attribution

Section 2 develops the higher-order Wick/transfer route; Section 3 gives the multiplication/energy route. They meet at the scalar moment theorem. Section 4 completes the common projected-moment and sample-label steps. Section 5 computes the exact benchmark, and Section 6 compares constants and numerical certificates. The original rMPS comparison is followed by a separate mixed reconstruction application: the diagram on page 19 maps Chain C, and Sections 8–13 supply its proof. This is an application of the combined Wick–Hermite organization, not a third independent proof of the same rMPS endpoint. Appendix D records the attribution boundaries across both applications.

The two routes have identifiable public antecedents. Rakhshan–Rabusseau [2, Theorem 1 and Section 5.1, Lemmas 3–4] analyze Gaussian tensor-train projections using a fourth-order one-core Frobenius contraction and its iteration along the tensor train. This is the direct predecessor of the conditional-energy calculation in Chain B. Its extension to every integer moment order is proved below; the one-core strategy itself is not claimed as new.

Camaño–Epperly–Meyer–Tropp [1, Section 4.4.1, equation (4.8), and Theorems 4.3–4.4] obtain the exact fourth-moment tensor and its fractured-Gaussian surrogate; Section 4.6 treats the complex case. Chain A extends their original-core contraction methodology to the higher orders needed here. That extension is not attributed to a published eighth-order theorem in their paper.

The local Wick identities are classical [3, 4]; the transfer kernels are normalized orthogonal/unitary pairing Gram matrices [5, Section 2]. Chain B’s multiplication and energy interfaces follow the supplied v11 manuscript [12], while its Gaussian conditioning inputs and the sample-label expansion have independent classical proofs. The contribution of this note is the explicit higher-order calculation, the specified centered-Gram formulas, and their comparison, not origination of these ingredients. This attribution audit identifies directly relevant antecedents; it is not an exhaustive publication-priority determination.

2 Chain A: the higher-order Wick/transfer proof

The real eighth moment has 105 local pairing states; the complex eighth moment has 24 permutation states. Their transfer matrices have constant row sums that can be evaluated exactly. No norm estimate of a large coefficient tensor is needed to sum those weights.

2.1 A1. Integrate repeated occurrences of a core jointly

For a real centered Gaussian family, the Wick–Isserlis rule writes an even moment as a sum over pairings, with each pair contributing its covariance [3]. For independent circular complex Gaussians, applying that rule to real and imaginary parts leaves only pairings of an unconjugated occurrence with a conjugated occurrence; balanced moments are indexed by permutations. Goodman [4] supplies the classical complex Gaussian/Wishart setting. These Gaussian moment identities are not new ingredients.

Expanding $|u^*\omega|^{2p}$ produces $2p$ occurrences of each original core. Integrating that core once, with all its occurrences present, yields the local states

$$\mathcal{P}_2(2p) \text{ over } \mathbb{R}, \quad S_p \text{ over } \mathbb{C}.$$

Their cardinalities are $(2p - 1)!!$ and $p!$, respectively. No repeated core is replaced by a fresh independent copy.

2.2 A2. Form the bond transfer matrix

For real pairings $\alpha, \beta \in \mathcal{P}_2(2p)$, let $\ell(\alpha, \beta)$ be the number of components in their union. A pair shared by both matchings is counted as one component. Each component leaves one freely summable bond index, contributing a factor χ . The normalization from the $2p$ copies contributes χ^{-p} per bond. Thus

$$T_p(\alpha, \beta) = \chi^{\ell(\alpha, \beta) - p}. \quad (2.1)$$

In the complex model the analogous transfer is

$$T_p^{\mathbb{C}}(\sigma, \tau) = \chi^{\text{cyc}(\sigma^{-1}\tau) - p}, \quad \sigma, \tau \in S_p, \quad (2.2)$$

where cyc counts permutation cycles.

At $p = 2$, the real formula reduces to

$$T_2 = \begin{pmatrix} 1 & \chi^{-1} & \chi^{-1} \\ \chi^{-1} & 1 & \chi^{-1} \\ \chi^{-1} & \chi^{-1} & 1 \end{pmatrix}. \quad (2.3)$$

This is precisely the fourth-order bond transfer in [1, Section 4.4.1, equation (4.8)]; (2.1) retains the additional pairing states required at higher orders.

As matrices, (2.1)–(2.2) are χ^{-p} times the standard orthogonal and unitary pairing Gram matrices; see [5, Section 2]. The real Gram entry is χ^ℓ , and restricting to balanced pairings gives the permutation cycle formula. We use these *Gram matrices*, not their Weingarten pseudoinverses. No Haar-distributed core or replacement of the Gaussian law is involved. Their occurrence in the present higher-order rMPS contraction is derived here; the underlying Gram kernels are not new.

2.3 A3. Evaluate the row sum without enumerating every path

The following elementary recurrences evaluate a row sum of the classical Gram kernels just identified. They are included to make the constants explicit, not as new combinatorial identities.

Lemma 2.1 (Bond-transfer row sum). *For every integer $p \geq 1$, the transfer matrix in the corresponding field obeys*

$$T_p \mathbf{1} = \rho_p(\chi) \mathbf{1}. \quad (2.4)$$

Proof. For a fixed real pairing α , define

$$F_p(x) = \sum_{\beta \in \mathcal{P}_2(2p)} x^{\ell(\alpha, \beta)}.$$

Relabeling positions shows that this does not depend on α . Remove a designated pair $\{u, v\}$ of α . If β also pairs u to v , the removed component contributes a factor x . Otherwise β contains $\{u, a\}$ and $\{v, b\}$. Delete u, v and replace those two pairs by $\{a, b\}$. For each smaller pairing there are $p-1$ choices of $\{a, b\}$ and two choices of its orientation; the number of components is unchanged. Hence

$$F_p(x) = (x + 2p - 2)F_{p-1}(x), \quad F_1(x) = x.$$

It follows that

$$F_p(\chi) = \chi(\chi + 2) \cdots (\chi + 2p - 2).$$

Divide by χ^p to obtain (2.4) for real cores.

For complex cores, multiplication by a fixed permutation reduces a row sum to $\chi^{-p} \sum_{\pi \in S_p} \chi^{\text{cyc}(\pi)}$. When a new symbol is added to a permutation, it either forms a singleton cycle or is inserted immediately after one of the $p-1$ existing symbols. Thus the cycle-count polynomial obeys the recurrence with factor $x + p - 1$, giving

$$\sum_{\pi \in S_p} x^{\text{cyc}(\pi)} = x(x + 1) \cdots (x + p - 1).$$

This proves the complex formula. □

At $p = 4$, the two row sums are

$$\rho_4(\chi) = 1 + \frac{12}{\chi} + \frac{44}{\chi^2} + \frac{48}{\chi^3} = \left(1 + \frac{2}{\chi}\right) \left(1 + \frac{4}{\chi}\right) \left(1 + \frac{6}{\chi}\right), \quad \mathbb{F} = \mathbb{R}, \quad (2.5)$$

$$\rho_4(\chi) = 1 + \frac{6}{\chi} + \frac{11}{\chi^2} + \frac{6}{\chi^3} = \left(1 + \frac{1}{\chi}\right) \left(1 + \frac{2}{\chi}\right) \left(1 + \frac{3}{\chi}\right), \quad \mathbb{F} = \mathbb{C}. \quad (2.6)$$

The real coefficients sum to 105, and the complex coefficients sum to 24, as they should when $\chi = 1$.

2.4 A4. Control arbitrary coefficient tensors and propagate

Lemma 2.2 (Coefficient-network bound). *For any fixed choice of the local Wick states, the contraction D_u of the $2p$ coefficient tensors satisfies*

$$|D_u| \leq \|u\|^{2p}. \quad (2.7)$$

Proof. Each physical index is joined to exactly one other occurrence by a pairing. The coefficient network therefore has one vertex per copy of u or $\bar{u}$, edges joining pairs of vertices, and no self-loops. For two tensors contracted over all their shared indices, reshape the shared indices into a single matrix index. Matrix multiplication and Frobenius Cauchy–Schwarz give

$$\|\text{contract}(A, B)\|_F \leq \|A\|_F \|B\|_F.$$

Merge two vertices, contracting all edges between them, and repeat. At every step the resulting boundary tensor is retained in full; contractions inside the merged cluster have already been performed. Each connected component ends as a scalar bounded by the product of its initial Frobenius norms. Multiplying these bounds proves (2.7). Complex conjugation does not change the norm. No assumption about coefficient signs is used. $\square$

Let $c_{\mathbb{R},p} = (2p-1)!!$ and $c_{\mathbb{C},p} = p!$. The Wick expansion in the real case is

$$\mathbb{E}|u^* \omega|^{2p} = \sum_{\alpha_1, \dots, \alpha_n} \left[\prod_{j=1}^{n-1} T_p(\alpha_j, \alpha_{j+1}) \right] D_u(\alpha_1, \dots, \alpha_n). \quad (2.8)$$

The complex expansion has the same structure with permutation labels. The weights are nonnegative, so Lemmas 2.1 and 2.2 give the common scalar endpoint:

Proposition 2.3 (Sharp scalar moments). *For every $u \in \mathbb{F}^{d^n}$ and integer $p \geq 1$,*

$$\boxed{\mathbb{E}|u^* \omega|^{2p} \leq c_{\mathbb{F},p} \mu_p \|u\|^{2p}}. \quad (2.9)$$

Equality holds for product tensors.

Completion of Chain A. Equation (2.8) is at most

$$\|u\|^{2p} \mathbf{1}^\top T_p^{n-1} \mathbf{1} = c_{\mathbb{F},p} \rho_p(\chi)^{n-1} \|u\|^{2p}.$$

For $u = u_1 \otimes \dots \otimes u_n$, each physical-site contraction is a product of pairwise inner products of identical vectors (or of conjugate pairs). Hence every D_u equals $\|u\|^{2p}$, and no inequality is lost. $\square$

The 105 and 24 local labels at eighth order do not imply that arbitrary coefficient-network evaluation is computationally cheap. For this uniform bound, exact evaluation is unnecessary: (2.7) handles the coefficients, and (2.4) sums the transfer weights.

3 Chain B: original-law multiplication and conditional energy

The same moment factor follows from a single Gaussian matrix inequality. The multiplication and exact-cutoff interfaces locate the relevant original-law energy; conditional integration evaluates it without enumerating Wick states. This route can also be written as a standalone conditioning proof.

The fourth-order version of the one-core contraction and its tensor-train iteration are already in [2, Section 5.1, Lemmas 3–4]. We generalize that local estimate in moment order and place the resulting calculation inside the original-law interfaces of [12]. This is not a claim that the conditional-contraction strategy originated in the framework.

3.1 B1. Start from the original-law vacuum identity

Let γ be the joint law of the original real Gaussian coordinates in all cores and probes. In the complex case, use the independent real and imaginary coordinates with their $1/\sqrt{2}$ normalization. On $\mathcal{H} = \mathbb{F}^2 \otimes L^2(\gamma)$, let $\Omega = 1$ and let $\mathcal{M}_G$ be multiplication by the Hermitian matrix G . Then

$$\mathbb{E} \operatorname{tr} G^4 = \sum_{b=1}^2 \|\mathcal{M}_G^2(e_b \otimes \Omega)\|^2. \quad (3.1)$$

This follows directly from $\mathcal{M}_G^j(e_b \otimes \Omega) = G^j e_b$ and $\sum_b \|G^2 e_b\|^2 = \operatorname{tr} G^4$. The dimension factor comes from the external space $\mathbb{F}^2$, not from the polynomial state space.

3.2 B2. Use an exact local-degree and active-probe cutoff

Give the Gaussian polynomial space its Hermite multigrading by the original cores; the underlying Gaussian chaos and tensor-product structure is classical [8, Chapters 2–4]. An entry of $y_a y_a^*$ has degree at most two in each core of probe a . After $j \leq 2$ multiplications, the local degree in every core is at most $2j$, and a Hermite term has at most j active probe blocks. A block is active when its local component is nonconstant; this is not the same as having appeared earlier in an expanded word.

Let Π restrict to local core degrees at most four and at most two active probe blocks. It may also impose the exact invariance under multiplying any whole core by -1 , which fixes G and the vacuum. Set

$$M = \Pi \mathcal{M}_G \Pi.$$

On this finite-dimensional space, multiplication is well defined and

$$M^j(e_b \otimes \Omega) = \mathcal{M}_G^j(e_b \otimes \Omega), \quad j = 0, 1, 2. \quad (3.2)$$

Thus

$$\mathbb{E} \operatorname{tr} G^4 = \sum_{b=1}^2 \|M^2(e_b \otimes \Omega)\|^2. \quad (3.3)$$

This is the exact finite-moment compression interface of v11 [12]. Taking the squared norm involves degree at most eight in each core, which explains why eighth-order probe information is needed. No large sample outcome is clipped, and no full-cutoff operator-norm bound is invoked.

3.3 B3. Retain the coefficient Gram across one core

In the real case at $p = 2$, after transposing the matrix product and matching the variance normalization, [2, Lemma 4] gives exactly

$$\mathbb{E}_R \|RC\|_{\mathbb{F}}^4 = \|C\|_{\mathbb{F}}^4 + \frac{2}{\chi} \operatorname{tr}((CC^*)^2) \leq \left(1 + \frac{2}{\chi}\right) \|C\|_{\mathbb{F}}^4.$$

Their proof uses standard Wishart moments. The next proof uses the same Gaussian Gram geometry and convexity to cover all integer orders and both fields.

Lemma 3.1 (One-core Gaussian energy inequality). *Let $C \in \mathbb{F}^{s \times t}$ be deterministic, and let $R \in \mathbb{F}^{\chi \times s}$ have independent Gaussian entries of variance $1/\chi$. For every integer $p \geq 1$,*

$$\boxed{\mathbb{E}_R \|RC\|_{\mathbb{F}}^{2p} \leq \rho_p(\chi) \|C\|_{\mathbb{F}}^{2p}.} \quad (3.4)$$

Equality holds when C has rank at most one.

Proof. Let $\lambda_1, \dots, \lambda_k$ be the nonzero eigenvalues of CC^* . Gaussian orthogonal or unitary invariance gives

$$\|RC\|_{\mathbb{F}}^2 \stackrel{d}{=} \sum_{j=1}^k \lambda_j Q_j, \quad Q_j \stackrel{d}{=} \frac{1}{\chi} \sum_{\ell=1}^{\chi} |g_{\ell}|^2,$$

where the Q_j are independent. The gamma density, or its elementary integration-by-parts recursion, gives

$$\mathbb{E} Q_j^p = \prod_{l=0}^{p-1} \left(1 + \frac{\eta^l}{\chi}\right) = \rho_p(\chi). \quad (3.5)$$

For real cores, Q_j is χ^{-1} times a chi-square variable with χ degrees of freedom. For complex cores it has gamma shape χ and rate χ .

If $s_0 = \sum_j \lambda_j = 0$, the assertion is trivial. Otherwise convexity yields

$$\left(\sum_j \frac{\lambda_j}{s_0} Q_j \right)^p \leq \sum_j \frac{\lambda_j}{s_0} Q_j^p.$$

Take expectations and multiply by s_0^p to obtain (3.4). With only one nonzero eigenvalue the inequality is an equality. $\square$

Positivity is used only for the eigenvalues of the Gram matrix CC^* , not for the entries of C . The estimate therefore applies to arbitrary signed or complex coefficient tensors. It does not require a positive polynomial cone.

3.4 B4. Iterate the conditional energy and close the scalar bound

This is the independent-core/Frobenius-reshaping iteration already used at fourth order in [2, Section 5.1], now with Lemma 3.1 at order $2p$. We give the full argument to specify normalization and equality cases.

Independent proof of Proposition 2.3. Start with the coefficient tensor $\bar{u}$. Reshape its first physical index as a row index and its remaining physical indices as columns. Its Frobenius norm is $\|u\|$.

Scale the first $n-1$ cores by $\chi^{-1/2}$. Contracting the first core is exactly the operation $C \mapsto RC$ in Lemma 3.1. Before the next contraction, combine the current bond index and the next physical index into the row index. This reshaping preserves the Frobenius norm; all unprocessed physical indices remain in the column index.

Write C_j for the retained coefficient matrix after the first j cores and $S_j = \|C_j\|_{\mathbb{F}}^2$, with $S_0 = \|u\|^2$. The next core is independent of the already processed ones, so

$$\mathbb{E}[S_j^p \mid C_{j-1}] \leq \rho_p(\chi) S_{j-1}^p, \quad j = 1, \dots, n-1. \quad (3.6)$$

Iterating conditional expectations gives

$$\mathbb{E} S_{n-1}^p \leq \rho_p(\chi)^{n-1} \|u\|^{2p}.$$

The last unscaled core makes a scalar Gaussian linear form with conditional variance S_{n-1} . Its $2p$ -th absolute moment is $c_{\mathbb{F},p} S_{n-1}^p$. Taking expectation proves (2.9).

For a product tensor, every intermediate coefficient matrix has rank at most one. All uses of (3.4) are then equalities, proving sharpness. $\square$

Each original core is integrated jointly with all its repeated occurrences. The independence used in (3.6) is independence of the *next core* from the preceding ones, not independence of repeated uses of one core. Only the retained coefficient boundary is reshaped.

The proof of (2.9) works at all integer orders. For P10 only $p \leq 4$ is required. The local-degree cutoff identifies the vacuum energy needed for the centered matrix calculation, while the conditional contraction supplies its bond-sensitive moment estimates. The two interfaces are compatible, but neither their use here nor their notation establishes that this fixed-order argument requires the specialized graded machinery.

4 Shared completion: projected moments and ordered sample energy

The two chains have already converged at Proposition 2.3. The remaining proof is common: convert scalar estimates into projected-vector moments and expand the centered sample word. Keeping the alternating pairing is necessary for the exact constants.

4.1 From scalar projections to the projected norm

Lemma 4.1 (Projected radial moments). *For any deterministic isometry $U : \mathbb{F}^r \rightarrow \mathbb{F}^{d^n}$ and integer $p \geq 1$,*

$$\mathbb{E} \|U^* \omega\|^{2p} \leq d_{r,p} \mu_p, \quad d_{r,p} = \prod_{j=0}^{p-1} (r + \eta j). \quad (4.1)$$

For $r \leq d$, the product-prefix subspace with r last-site directions attains equality.

Proof. Let θ be uniform on the unit sphere in $\mathbb{F}^r$. Gaussian polar decomposition gives, for deterministic v ,

$$\mathbb{E}_\theta |\theta^* v|^{2p} = \frac{c_{\mathbb{F},p}}{d_{r,p}} \|v\|^{2p}.$$

Indeed, a standard Gaussian vector is its independent radius times its direction; the radius has $2p$ -th moment $d_{r,p}$ and any unit scalar projection has moment $c_{\mathbb{F},p}$. Apply Proposition 2.3 to $U\theta$, whose norm is one, and use Fubini. Equality holds on the product-prefix family because every $U\theta$ is a product tensor. $\square$

For $r = 2$, writing $R = \|y\|^2$, this gives

	$\mathbb{E} R^2$	$\mathbb{E} R^4$
$\mathbb{F} = \mathbb{R}$	$\leq 8\mu_2$	$\leq 384\mu_4$
$\mathbb{F} = \mathbb{C}$	$\leq 6\mu_2$	$\leq 120\mu_4$

(4.2)

The scalar eighth-moment bounds behind the right column are $105\mu_4\|u\|^8$ and $24\mu_4\|u\|^8$, respectively. These are original eighth moments, not consequences of matching only four moments.

4.2 The exact fourth moment of a centered sample average

The sample-label bookkeeping is the entrywise moment–cumulant principle for independent sums; see [7, Chapter 2]. The following ordered-matrix specialization follows by direct expansion and is not claimed as a new general identity. The matrix order must be retained even though each entry is scalar.

Lemma 4.2 (Ordered fourth-order sample assembly). *Let $X_1, \dots, X_N$ be independent identically distributed centered Hermitian matrices with finite fourth moments. Write $V = \mathbb{E}X_1^2$ and let X' be an independent copy. Then*

$$\mathbb{E} \operatorname{tr} \left(\frac{1}{N} \sum_{a=1}^N X_a \right)^4 = \frac{\mathbb{E} \operatorname{tr} X_1^4 + (N-1)[2 \operatorname{tr} V^2 + \mathbb{E} \operatorname{tr}(X_1 X' X_1 X')]}{N^3}. \quad (4.3)$$

Moreover,

$$|\mathbb{E} \operatorname{tr}(X_1 X' X_1 X')| \leq \operatorname{tr} V^2. \quad (4.4)$$

Proof. Expand the ordered word of length four in its sample labels. A label appearing exactly once gives zero after conditioning on the other matrices. There are N words with one label four times. For each unordered pair of distinct labels, there are six words: four are cyclically equivalent to $X_1^2(X')^2$, and two alternate. The former have expected trace $\operatorname{tr} V^2$. Multiplication by $\binom{N}{2}$ and division by N^4 give (4.3).

For Hermitian A, B , Hilbert–Schmidt Cauchy–Schwarz gives

$$|\operatorname{tr}(ABAB)| = |\langle BA, AB \rangle_{\text{HS}}| \leq \|BA\|_{\text{F}} \|AB\|_{\text{F}} = \operatorname{tr}(A^2 B^2).$$

Take expectations and use independence to obtain (4.4). $\square$

The lemma is also the $Q(T) = T^2$ instance of v11’s fourth-order original-law energy identity with component reuse [12]. If K is a centered Gaussian Hermitian matrix with the complete real-entry covariance of X_1 (the joint real/imaginary covariance in the complex model), then

$$\mathbb{E} \operatorname{tr} G^4 = \frac{N-1}{N^3} \mathbb{E} \operatorname{tr} K^4 + \frac{1}{N^3} \mathbb{E} \operatorname{tr} X_1^4. \quad (4.5)$$

Here $\mathbb{E} \operatorname{tr} K^4 = 2 \operatorname{tr} V^2 + \mathbb{E} \operatorname{tr}(X_1 X' X_1 X')$. This is a comparison of the centered matrices X_a , not a substitution of Gaussian surrogates for the original tensor cores.

4.3 Proof of the uniform endpoint

Proof of Theorem 1.1. For $Y = yy^*$, $X = Y - I_2$, and $R = \|y\|^2$, the eigenvalues of X are $R - 1$ and -1 . Since isotropy gives $\mathbb{E}R = 2$,

$$\mathbb{E} \operatorname{tr} X^4 = \mathbb{E}R^4 - 4\mathbb{E}R^3 + 6\mathbb{E}R^2 - 6.$$

With an independent copy R' , the inequality $\mathbb{E}[(R^2 - (R')^2)(R - R')] \geq 0$ implies $\mathbb{E}R^3 \geq (\mathbb{E}R^2)(\mathbb{E}R) = 2\mathbb{E}R^2$. Hence

$$\mathbb{E} \operatorname{tr} X^4 \leq \mathbb{E}R^4. \quad (4.6)$$

Also $V = \mathbb{E}X^2 \succeq 0$, so

$$\operatorname{tr} V^2 \leq (\operatorname{tr} V)^2 = (\mathbb{E}R^2 - 2)^2. \quad (4.7)$$

Substitute (4.2), (4.4), (4.6), and (4.7) into (4.3). This gives (1.5) and (1.6) with exactly the constants stated. $\square$

Neither scalar route requires independence of the coordinates of y . The sample-label identity uses only independence of different probes, while their within-probe correlations are retained in the moments of X .

5 The common exact product-prefix calculation

For the specified subspace U_0 , the projected vector has the exact original-law representation $y \stackrel{d}{=} \sqrt{W} z$. Both coordinates share W . This closes the radial moment boundary and permits exact evaluation of the alternating sample word, not only a bound on it.

5.1 A closed radial boundary from the original cores

For U_0 in (1.7), define the prefix row states

$$v_1 = \chi^{-1/2} H_1(1, :), \quad v_j = \chi^{-1/2} v_{j-1} H_j(:, 1, :), \quad 2 \leq j \leq n-1.$$

The final output is

$$y = (v_{n-1} H_n(:, 1), v_{n-1} H_n(:, 2))^T.$$

Conditionally on preceding cores, the coordinates of v_j are independent Gaussians with variance $\|v_{j-1}\|^2/\chi$. Thus the ratio $\|v_j\|^2/\|v_{j-1}\|^2$ has the law of Q in (3.5), independent of the preceding sigma-field. The denominator is nonzero almost surely. The first factor has the same law, and the last core supplies a normalized Gaussian vector independent of these ratios. Consequently

$$\boxed{y \stackrel{d}{=} \sqrt{W} z, \quad W = \prod_{j=1}^{n-1} Q_j, \quad z \sim N_{\mathbb{F}}(0, I_2),} \quad (5.1)$$

where $Q_1, \dots, Q_{n-1}, z$ are independent and $\mathbb{E}W^p = \mu_p$. This is derived from the full core law. It is not a decoupling assumption.

On moment functions through order four, the radial conditional-integration operator has the exact update

$$(\mathbb{T}_\chi f)(s) = \mathbb{E}f(sQ), \quad \mathbb{T}_\chi(s^p) = \rho_p(\chi)s^p. \quad (5.2)$$

Its matrix in the basis $1, s, s^2, s^3, s^4$ is $\text{diag}(1, 1, \rho_2, \rho_3, \rho_4)$. This is a five-dimensional closure for these moment functions on the prefix family, not for the full probability law or arbitrary tensor coefficients. The Wick route obtains the same moments because every coefficient contraction for a product direction attains equality.

5.2 The one-probe fourth moment

Let $s = \|z\|^2$, so $R = Ws$ and

$$\mathbb{E}s^p = \prod_{j=0}^{p-1} (2 + \eta j).$$

For $a = \mu_2$, $b = \mu_3$, $c = \mu_4$, direct expansion gives

$$A_{\mathbb{F}} := \mathbb{E} \text{tr} X^4 = \mathbb{E}R^4 - 4\mathbb{E}R^3 + 6\mathbb{E}R^2 - 6, \quad (5.3)$$

$$V := \mathbb{E}X^2 = vI_2, \quad v = (2 + \eta)a - 1. \quad (5.4)$$

For the real field, $(\mathbb{E}s^2, \mathbb{E}s^3, \mathbb{E}s^4) = (8, 48, 384)$; for the complex field they are $(6, 24, 120)$. Substitution proves the values of $A_{\mathbb{F}}$ in (1.9)–(1.10), and $\mathbb{E} \text{tr} G^2 = 2v/N$ proves (1.11).

5.3 The alternating word and the exact pair coefficient

For deterministic real symmetric or complex Hermitian D , Gaussian fourth-moment contraction yields

$$\mathbb{E}[zz^* D z z^*] = \eta D + (\text{tr } D)I_2.$$

Expanding $XD X = (W z z^* - I_2)D(W z z^* - I_2)$ therefore gives

$$\mathcal{L}(D) := \mathbb{E}[XD X] = (\eta a - 1)D + a(\text{tr } D)I_2. \quad (5.5)$$

In addition,

$$\mathbb{E} \text{tr } X^2 = 2v, \quad \mathbb{E}(\text{tr } X)^2 = 2(2 + \eta)a - 4.$$

Condition on X and integrate X' to obtain

$$\begin{aligned} C_{\mathbb{F}} &:= \mathbb{E} \text{tr}(X X' X X') = \mathbb{E} \text{tr}(X \mathcal{L}(X)) \\ &= (\eta a - 1)2v + a(2(2 + \eta)a - 4). \end{aligned} \quad (5.6)$$

Explicitly,

$$C_{\mathbb{R}} = 24a^2 - 16a + 2, \quad C_{\mathbb{C}} = 12a^2 - 12a + 2.$$

Since $2 \text{tr } V^2 = 4v^2$, the pair coefficient is

$$B_{\mathbb{F}} = 4v^2 + C_{\mathbb{F}}.$$

This gives $B_{\mathbb{R}} = 88a^2 - 48a + 6$ and $B_{\mathbb{C}} = 48a^2 - 36a + 6$. Inserting these and (5.3) into Lemma 4.2 completes the proof of Theorem 1.2.

5.4 A normalization check

As $\chi \rightarrow \infty$ with n fixed, $\mu_p \rightarrow 1$. Both chains therefore give

$$\mathbb{E} \text{tr } G^4 \longrightarrow \begin{cases} 46/N^2 + 188/N^3, & \mathbb{F} = \mathbb{R}, \\ 18/N^2 + 36/N^3, & \mathbb{F} = \mathbb{C}. \end{cases} \quad (5.7)$$

The one-probe and pair constants in this limit are $(234, 46)$ and $(54, 18)$, respectively. These follow by direct substitution, without a limiting interchange of unbounded random variables.

6 End results and explicit constants

6.1 The same constants, not merely the same order

Quantity	Chain A: Wick / transfer	Chain B: multiplication / energy
One-bond moment factor	$\rho_p(\chi)$ from the exact row sum	$\rho_p(\chi)$ from conditional energy
Length multiplier	$\mu_p = \rho_p(\chi)^{n-1}$	$\mu_p = \rho_p(\chi)^{n-1}$
Real scalar eighth-moment allowance for $\ u\ = 1$	$105\mu_4$	$105\mu_4$
Complex scalar eighth-moment allowance for $\ u\ = 1$	$24\mu_4$	$24\mu_4$
Projected real eighth-moment allowance, dimension two	$384\mu_4$	$384\mu_4$
Projected complex eighth-moment allowance, dimension two	$120\mu_4$	$120\mu_4$
Uniform Gram fourth moment	(1.5) and (1.6)	Exactly the same bounds
Product-prefix Gram fourth moment	(1.8)–(1.10)	Exactly the same formula

The scalar and projected radial bounds are sharp because product directions or the product-prefix subspace attain them. The *uniform centered Gram* bound need not be sharp: its common completion uses $\text{tr } V^2 \leq (\text{tr } V)^2$, bounds the alternating word in absolute value, and replaces $\mathbb{E} \text{tr } X^4$ by $\mathbb{E} \|y\|^8$. These losses are shared by both chains.

6.2 Why the constants coincide: the same rank-one Gaussian extremum

Equal constants do not follow merely from studying the same random law: two estimates of that law could lose different amounts. Here both routes preserve the exact same local Gaussian moment and then take the same worst-case coefficient envelope. The extremizer belongs to the class under consideration.

The Wick row sum is a radial Gaussian moment.

Let $g \in \mathbb{F}^x$ be standard Gaussian and put $Q = \|g\|^2/\chi$. Expanding $(\sum_i g_i^2)^p$ in the real case fixes one pairing α that identifies the two appearances of each squared coordinate. Applying Isserlis' formula adds the second pairing β . Every component of $\alpha \cup \beta$ contributes one free coordinate index. Thus

$$\underbrace{\sum_{\beta \in \mathcal{P}_2(2p)} \chi^{\ell(\alpha, \beta) - p}}_{\text{Chain A: one row sum}} = \underbrace{\mathbb{E} \left(\frac{\|g\|^2}{\chi} \right)^p}_{\text{Chain B: gamma moment}} = \rho_p(\chi).$$

The complex identity uses balanced pairings, equivalently permutations, in place of $\mathcal{P}_2(2p)$. It has the same interpretation. This is the exact link between the two calculations, not an analogy between their notation. The Wick identities [3, 4] and the pairing Gram representation [5] give the two ways of reading this equality.

Both coefficient bounds are saturated by rank one.

For $K = CC^* \succeq 0$, let $s = \text{tr } K$ and $S = \|RC\|_{\mathbb{F}}^2$. If λ_j are the eigenvalues of K , then $S \stackrel{d}{=} \sum_j \lambda_j Q_j$. The convexity step in Chain B gives

$$\sup_{C: \|C\|_{\mathbb{F}}^2 = s} \mathbb{E} S^p = s^p \rho_p(\chi),$$

where any admissible rank-one C attains the supremum. For a product coefficient tensor, every intermediate matrix has rank one, while every coefficient network in Chain A attains its Frobenius bound. Equality therefore persists through all $n - 1$ bonds in both proofs. The scalar factor $c_{\mathbb{F},p}\mu_p$ cannot be lowered in a theorem uniform over all u .

A trace-power calculation exposes the common information loss.

This also follows from the classical trace-cumulant description of Gaussian quadratic forms [6]; the complex setting is treated in [4]. Write $\tau_k = \text{tr } K^k$, so $\tau_1 = s$, and set $\gamma_\chi = \eta/\chi$. Diagonalization and the gamma moment-generating function give, for t near zero,

$$\mathbb{E}e^{tS} = \det(I - \gamma_\chi tK)^{-1/\gamma_\chi}, \quad \kappa_k(S) = (k-1)!\gamma_\chi^{k-1}\tau_k.$$

The second formula is obtained by expanding the logarithm of the first; no new quadratic-form cumulant theorem is needed. Substituting these cumulants into the fourth moment-cumulant identity [7, Chapter 2] yields

$$\mathbb{E}S^4 = s^4 + 6\gamma_\chi s^2\tau_2 + 3\gamma_\chi^2\tau_2^2 + 8\gamma_\chi^2s\tau_3 + 6\gamma_\chi^3\tau_4.$$

All coefficients are nonnegative. Replacing τ_k by s^k gives

$$\mathbb{E}S^4 \leq s^4(1 + 6\gamma_\chi + 11\gamma_\chi^2 + 6\gamma_\chi^3) = s^4\rho_4(\chi).$$

At rank one, $\tau_k = s^k$, so the bound is exact. With $\eta = 2$ this is the Wick row polynomial $(1, 12, 44, 48)$, and with $\eta = 1$ it is $(1, 6, 11, 6)$; compare (2.5)–(2.6). The loop expansion and the spectral-energy calculation thus collapse the same positive Gaussian contraction to the same rank-one envelope.

The equality continues because the rest of the proof is shared.

Spherical averaging contributes the same exact factor $d_{r,p}/c_{\mathbb{F},p}$. The ordered sample-label identity is exact, and the subsequent three upper bounds described above are the same in both chains. The equality of the constants tables follows from this common completion, not from a theorem that all Wick and graded proofs must have equal constants.

There is also an operator interpretation. The symmetric nonnegative transfer has $T_p\mathbf{1} = \rho_p\mathbf{1}$ and operator norm ρ_p . On the rank-one boundary, conditional integration sends s^p to $\mathbb{E}(sQ)^p = \rho_p s^p$. These are two representations of the same invariant scalar moment. For arbitrary coefficient matrices, the radial description is an upper envelope, not an exact compression of their full law or an equivalence of the full operators.

What would be needed to improve a bound.

With a spread-out coefficient Gram, $\text{tr } K^2 < s^2$ and some higher trace powers are also strictly smaller than their rank-one values. Retaining these invariants can improve a coefficient-sensitive estimate in either language. A useful multi-site refinement must also justify how those invariants are updated after the next core contraction and reshaping. The worst-case scalar theorem cannot improve uniformly, because product tensors attain it; the uniform centered-Gram bound may still improve by avoiding its separate shared relaxations.

6.3 Tensor length versus bond dimension

Put $\tau = (n-1)/\chi$. The inequality $\log(1+x) \leq x$ gives

$$\mu_p \leq \exp\left(\frac{\eta p(p-1)}{2}\tau\right). \quad (6.1)$$

Consequently, both uniform theorems imply

$$\mathbb{E} \operatorname{tr} G^4 \leq \frac{192e^{4\tau}}{N^2} + \frac{384e^{12\tau}}{N^3}, \quad \mathbb{F} = \mathbb{R}, \quad (6.2)$$

$$\mathbb{E} \operatorname{tr} G^4 \leq \frac{108e^{2\tau}}{N^2} + \frac{120e^{6\tau}}{N^3}, \quad \mathbb{F} = \mathbb{C}. \quad (6.3)$$

For the product-prefix family the leading coefficients improve:

$$\mathbb{E} \operatorname{tr} G^4 \leq \frac{88e^{4\tau}}{N^2} + \frac{384e^{12\tau}}{N^3}, \quad \mathbb{F} = \mathbb{R}, \quad (6.4)$$

$$\mathbb{E} \operatorname{tr} G^4 \leq \frac{48e^{2\tau}}{N^2} + \frac{120e^{6\tau}}{N^3}, \quad \mathbb{F} = \mathbb{C}. \quad (6.5)$$

Here $\mu_3 \geq \mu_2 \geq 1$ implies $A_{\mathbb{R}} \leq 384\mu_4$ and $A_{\mathbb{C}} \leq 120\mu_4$, while $B_{\mathbb{R}} \leq 88\mu_2^2$ and $B_{\mathbb{C}} \leq 48\mu_2^2$. If χ is proportional to n , all these fixed-order constants remain bounded uniformly in the tensor length.

For comparison, a total-degree Gaussian hypercontractive estimate for the real scalar polynomial $u^*\omega$ gives

$$\|u^*\omega\|_{L_8} \leq 7^{n/2}\|u\|, \quad \mathbb{E}|u^*\omega|^8 \leq 7^{4n}\|u\|^8.$$

This is the degree- n consequence of Nelson's Gaussian hypercontractivity [9]; see also [8, Chapter 5] and the later reproof [10, equation (1.8) of the arXiv version]. The latter is a source for a proof, not the origin of the inequality. Both chains improve the bond-blind allowance to $105\mu_4 \leq 105e^{12\tau}$, so this gain is not a quantitative separation between Chains A and B.

6.4 The probability certificates being compared

The spectral theorem and Markov's inequality give

$$\mathbb{P}\{\|G\| > \varepsilon\} \leq \varepsilon^{-4} \mathbb{E} \operatorname{tr} G^4. \quad (6.6)$$

For the prefix family, define

$$(A, B) = (A_{\mathbb{F}}, B_{\mathbb{F}});$$

for the uniform bound, define

$$(A, B) = \begin{cases} (384\mu_4, 3(8\mu_2 - 2)^2), & \mathbb{F} = \mathbb{R}, \\ (120\mu_4, 3(6\mu_2 - 2)^2), & \mathbb{F} = \mathbb{C}. \end{cases} \quad (6.7)$$

In either case, the explicit sufficient certificate is

$$A + (N - 1)B \leq \delta\varepsilon^4 N^3. \quad (6.8)$$

A simpler, nonminimal sufficient condition is

$$N \geq \max \left\{ \left(\frac{2A}{\delta\varepsilon^4} \right)^{1/3}, \left(\frac{2B}{\delta\varepsilon^4} \right)^{1/2} \right\}. \quad (6.9)$$

This follows by bounding the fourth moment by $A/N^3 + B/N^2$. For fixed ε and τ , its leading high-confidence dependence is $\delta^{-1/2}$. This is not logarithmic confidence dependence or a claim of optimal sample complexity.

The second-moment prefix certificate used as a baseline is

$$N \geq \frac{h_{\mathbb{F}}\mu_2 - 2}{\delta\varepsilon^2}, \quad h_{\mathbb{R}} = 8, \quad h_{\mathbb{C}} = 6. \quad (6.10)$$

It concerns the *same* subspace U_0 . The comparison is between these specified moment certificates, not all algorithms or all concentration methods available for rMPS.

6.5 Concrete sufficient probe counts

All entries in Table 1 use $n = 100$, $\varepsilon = 1/2$, and $\delta = 0.01$. Each is the smallest integer satisfying the corresponding explicit certificate. The exact fourth-moment columns are repeated to make the absence of a difference between proof chains visible.

Field	χ	Second moment	Exact fourth moment		Uniform fourth
		Prefix U_0	Chain A	Chain B	Both chains
Real	100	21 929	4 611	4 611	5 234
Real	400	4 444	531	531	779
Real	1 000	3 100	360	360	540
Complex	100	5 628	706	706	1 009
Complex	400	2 274	253	253	396
Complex	1 000	1 850	201	201	322

Table 1: Sufficient probe counts from (6.10) and (6.8). The uniform column is valid for every fixed two-dimensional U . These are theorem certificates, not empirical thresholds.

Exact rational arithmetic checks both the passing integer and the preceding failing integer; Appendix A records the procedure. For real cores with $n = 100$, $\chi = 400$, $\varepsilon = 1/2$, and $\delta = 10^{-6}$, the same calculations give

Certificate	Sufficient N
Second moment, U_0	44 431 236
Exact fourth moment, U_0 (either chain)	51 182
Uniform fourth moment (either chain)	76 967

The improvement here is from using the fourth moment rather than the second, not from choosing Chain B instead of Chain A.

Comparison outcome. At every displayed parameter choice, the two fourth-order proof chains yield the same sufficient probe counts because they produce the same inequalities. Their difference is the retained description and proof organization, not a smaller numerical constant.

7 Scope, provenance, and what does not extend unchanged

7.1 A fourth-moment surrogate is not an eighth-moment identity

At fourth order, the real transfer has the special form

$$T_2 = \left(1 - \frac{1}{\chi}\right) I + \frac{1}{\chi} \mathbf{1}\mathbf{1}^\top.$$

This underlies the source paper’s fractured-Gaussian fourth-moment surrogate [1, Section 4.4.2, Theorem 4.4]. At eighth order, the transfer has distinct off-diagonal weights χ^{-1} , χ^{-2} , and χ^{-3} , and for $\chi > 1$ it is not the same two-term matrix form.

On a unit real product coordinate, the surrogate’s segmentation gives a product of $1 + \text{Binomial}(n - 1, 1/\chi)$ standard Gaussians. Its eighth moment is therefore

$$105 \left(1 + \frac{104}{\chi}\right)^{n-1}. \quad (7.1)$$

The actual rMPS eighth moment, computed by either chain, is

$$105 \left[\left(1 + \frac{2}{\chi}\right) \left(1 + \frac{4}{\chi}\right) \left(1 + \frac{6}{\chi}\right) \right]^{n-1}. \quad (7.2)$$

These generally differ. The precise limitation is that the existing *surrogate* cannot be reused unchanged. The original-core Wick method itself extends successfully, as Section 2 shows.

7.2 What the framework contributes in this comparison

Chain B uses the original-law multiplication identity, a moment-preserving reachable-state cutoff, a valid coefficient-boundary energy estimate, and the fourth-order sample energy assembly. It does not use the positive-coefficient cone theorem, connected-creation graph estimates, shared-factor occupation bounds, or covariance-spectrum renewal.

The one-core inequality can be proved independently by rotational invariance and convexity; its real fourth-order instance and tensor-train iteration have the direct published precedent [2, Section 5.1]. The centered-sample identity follows independently by expanding four labels, or by the classical moment-cumulant calculus [7]. The framework supplies a consistent organization and precise original-law interfaces, without presenting these Gaussian and cumulant identities as new. This pilot does not establish that its specialized operator machinery is indispensable or stronger than the extended Wick method.

The distinction is substantive: Chain B generalizes and reorganizes an existing Gaussian contraction strategy, and Chain A extends an existing Wick/transfer strategy. Section 6.2 identifies the shared rank-one extremum that explains their identical numerical constants.

7.3 The boundary of the result

The fixed-order P10 calculation supplies a uniform bound for arbitrary fixed two-dimensional subspaces and an exact formula on the specified coupled family. It does not supply a sharp theorem for growing subspace dimension, a prescribed optimal bond-dimension tradeoff, or logarithmic dependence on failure probability. Although the scalar bound holds at every integer order, the centered matrix assembly here has only been carried out at order four.

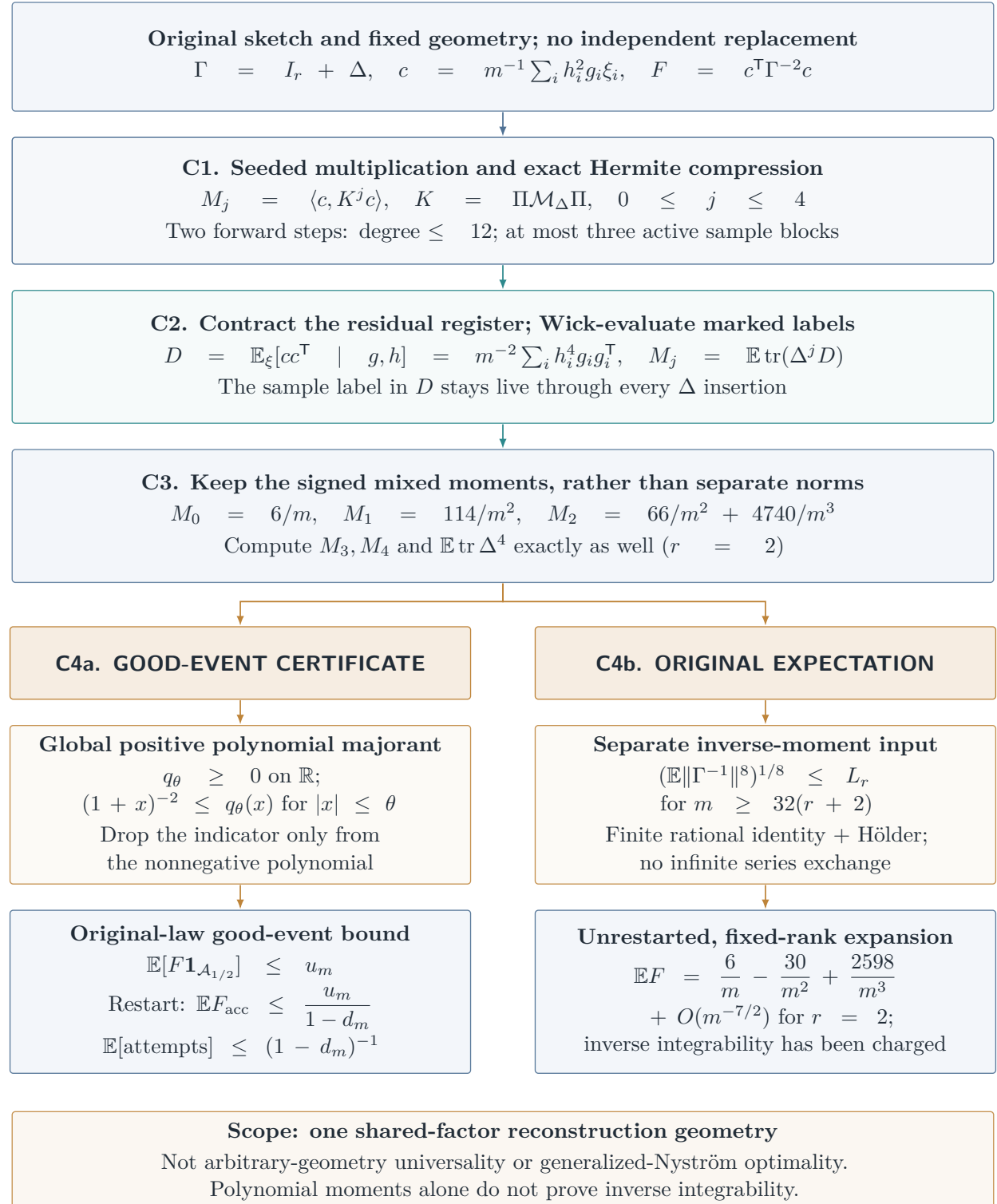
The note consolidates the rMPS proof and the subsequent two-method comparison from this project. It is an analytic derivation, not an independently reviewed proof, a formal proof-assistant certificate, or a publication-priority claim. Gaussian tensor-train projection and fourth-order conditional-contraction antecedents are credited to [2]; the exact fourth-moment rMPS transfer and surrogate analysis are credited to [1]. Standard pairing Gram matrices, Gaussian moment identities, and cumulant bookkeeping are credited at their points of use. The multiplication and energy interfaces are credited to [12], whose bibliographic author and title follow the supplied file. The higher-order comparison and P10 constants are proved here without asserting that no equivalent result exists elsewhere.

Retain the mixed reconstruction energy

Chain C: Wick contractions inside the original-law framework

Read downward. Blue retains the state; teal evaluates Gaussian moments.

Amber marks the two different analytic completions, not interchangeable claims.



8 Chain C: mixed reconstruction model and result

A new endpoint, not a third rMPS moment bound

The two rMPS chains above meet at a scalar moment bound. Chain C instead retains a *mixed reconstruction energy* involving both a random Gram inverse and a correlated residual coupling. For the rank-two, unit-residual model below, its original-estimator conclusion is

$$\mathbb{E}F = \frac{6}{m} - \frac{30}{m^2} + \frac{2598}{m^3} + O(m^{-7/2}).$$

The finite mixed moments and the good-event certificate do not require an inverse-tail estimate. The unconditional conclusion does, and that input is proved separately in Section 12.

8.1 Bridge from the rMPS radial boundary; local notation

The shared-factor row used here is a two-site, bond-one real rMPS row, up to swapping the two physical sites. Equivalently, it is the ordinary order-two Gaussian Khatri–Rao row already contained in the Gaussian tensor-train family [2, 1]. Restricting the first factor to $r + 1$ coordinates gives

$$(y, z) = h(g, \xi), \quad (y, z) \stackrel{d}{=} \sqrt{w}(g, \xi), \quad w = h^2.$$

This is the common radial boundary from Section 5, now with one direction reserved for the residual. The two occurrences of h are shared, not separately resampled. For equal physical dimensions take $d \geq r + 1$; extra coordinates are unused. The mixed polynomial formulas with general radial moments below can be reused for other radial laws, but the inverse-integrability theorem in this part is proved for $w = h^2$ only.

To avoid a collision with the *centered* rMPS matrix G in (1.2), this part denotes the reconstruction Gram by Γ and its centered error by $\Delta = \Gamma - I_r$. These are the quantities called G and E in the P02 problem statement. Here m is the sketch row count, r is the fixed subspace dimension, and $\mu_k = \mathbb{E}w^k = (2k - 1)!!$. There is no new assertion about arbitrary subspace–residual geometry or general rMPS bond dimension.

8.2 A specified orthogonal signal–residual geometry

Fix $r \geq 1$, $n_1 \geq r + 1$, $n_2 \geq 2$, and a unit coordinate vector $f_1 \in \mathbb{R}^{n_2}$. Define

$$u_a = e_a \otimes f_1 \quad (1 \leq a \leq r + 1), \quad Q = [u_1, \dots, u_r], \quad v = u_{r+1}.$$

Take, for example,

$$A = [Q, v], \quad B = [I_r, 0], \quad R = v e_{r+1}^\top,$$

so $A = QB + R$, $Q^\top R = 0$, and $\|R\|_F^2 = 1$. More generally the signal coefficient matrix is arbitrary and the rank-one residual can be scaled. All penalty formulas below scale by $\|R\|_F^2$.

Let $a_i \sim N(0, I_{n_1})$ and $b_i \sim N(0, I_{n_2})$ be independent over both factors and all i , and put

$$\Psi = \frac{1}{\sqrt{m}} \begin{pmatrix} (a_1 \otimes b_1)^\top \\ \vdots \\ (a_m \otimes b_m)^\top \end{pmatrix}.$$

Unused ambient coordinates play no role. The ambient dimensions may be arbitrarily large, including large enough that this is genuinely dimension-reducing. Write

$$g_i = (a_{i1}, \dots, a_{ir})^\top, \quad \xi_i = a_{i,r+1}, \quad h_i = b_{i1}, \quad w_i = h_i^2.$$

Here $g_i \sim N(0, I_r)$, $\xi_i, h_i \sim N(0, 1)$, and all these primitives are independent. Set

$$\Gamma = Q^\top \Psi^\top \Psi Q = \frac{1}{m} \sum_i w_i g_i g_i^\top, \quad \Delta = \Gamma - I_r, \quad c = \frac{1}{m} \sum_i w_i g_i \xi_i. \quad (8.1)$$

Then $C = Q^\top \Psi^\top \Psi R = c e_{r+1}^\top$. For $m \geq r$, Γ is positive definite almost surely: all h_i are nonzero almost surely and a Gaussian $m \times r$ matrix has full column rank.

Let $W = \Psi Q$ and reconstruct $\hat{A} = Q W^\dagger \Psi A$. Orthogonality gives the exact identity

$$\|A - \hat{A}\|_F^2 = 1 + F, \quad F = c^\top \Gamma^{-2} c. \quad (8.2)$$

Indeed $W^\dagger \Psi A = B + \Gamma^{-1} C$ and $R \perp \text{range } Q$. This fixed-range reconstruction is the established approximation $Q(\Psi Q)^\dagger \Psi A$, not a new algorithm. Tropp–Yurtsever–Udell–Cevher [16, equations (4.2)–(4.3), Lemmas A.3–A.4] give its orthogonal error decomposition and correction term; see also [13, Section 5.1, Lemma 5.1]. Their Remark 4.1 records still earlier algebraically equivalent approximations, so no algorithmic priority is assigned here. The fully iid real Gaussian benchmark is $\mathbb{E}F = r/(m - r - 1)$ for $m > r + 1$. It already appears for fixed-range reconstruction in [16, Lemma A.5]; Bartan–Pilanci [14, Lemma 1 in arXiv v2] give the corresponding regression prediction-error identity. Epperly–Webber [13, Theorem 2.1] explicitly cite Bartan–Pilanci for their real Gaussian benchmark. The standard Gaussian and pseudoinverse inputs are also in [15, Propositions 10.1–10.2]. Our Ψ is not an iid Gaussian matrix. Neither the error decomposition nor the Gaussian benchmark is claimed as a contribution of this note.

9 Mixed moments with the original sample labels

Define the original-law quantities

$$M_j = \mathbb{E}[c^\top \Delta^j c], \quad j \geq 0.$$

All occurrences in this expectation share the same variables in (8.1).

Theorem 9.1 (Rank-two mixed moments). *For $r = 2$ and every integer $m \geq 1$,*

$$\begin{aligned} M_0 &= \frac{6}{m}, & M_1 &= \frac{114}{m^2}, \\ M_2 &= \frac{66}{m^2} + \frac{4740}{m^3}, \\ M_3 &= \frac{5358}{m^3} + \frac{342756}{m^4}, \\ M_4 &= \frac{1962}{m^3} + \frac{508620}{m^4} + \frac{37984464}{m^5}. \end{aligned}$$

Also

$$\mathbb{E} \text{tr} \Delta^4 = \frac{654}{m^2} + \frac{36924}{m^3}. \quad (9.1)$$

These identities concern polynomials and remain valid even when $m < r$, where the inverse penalty is not defined as an ordinary inverse.

9.1 The live residual label

Condition on all g_i, h_i . Since the ξ_i 's are independent standard Gaussians,

$$\mathbb{E}_\xi[cc^\top \mid g, h] = D := \frac{1}{m^2} \sum_i w_i^2 g_i g_i^\top, \quad M_j = \mathbb{E} \text{tr}(\Delta^j D). \quad (9.2)$$

This is the ordinary conditional covariance calculation for a linear Gaussian response; it is not a new covariance identity. In particular $\mathbb{E}_\xi F = \text{tr}(\Gamma^{-1} D \Gamma^{-1})$ when the Gram is invertible. What the mixed calculation retains is the dependence of this covariance on the same design rows. It does not make D independent of Δ . In particular the same sample label in D may recur in every following factor of Δ .

9.2 First three moments with general radial weights

The next calculation works for any nonnegative iid w_i , independent of the Gaussians, with $\mathbb{E}w_i = 1$ and moments $\mu_k = \mathbb{E}w_i^k$. In our model $\mu_k = (2k - 1)!!$. Put $X_i = w_i g_i g_i^\top - I$ and $D_i = w_i^2 g_i g_i^\top$. Gaussian radial moments give

$$\mathbb{E}D_i = \mu_2 I, \quad \mathbb{E}X_i^2 = v_r I, \quad v_r = (r + 2)\mu_2 - 1.$$

The terms with a sample label appearing just once in an X -position vanish. Consequently

$$\begin{aligned} M_0 &= \frac{r\mu_2}{m}, \\ M_1 &= \frac{r[(r+2)\mu_3 - \mu_2]}{m^2}, \\ M_2 &= \frac{r}{m^3} \{ (m-1)\mu_2 v_r + (r+2)(r+4)\mu_4 - 2(r+2)\mu_3 + \mu_2 \}. \end{aligned} \quad (9.3)$$

For example, in M_2 , either the two X -labels agree and differ from the D -label, or all three labels agree. These are exactly the two terms in braces. Substitution of $\mu_2 = 3, \mu_3 = 15, \mu_4 = 105$ proves the first three lines of Theorem 9.1.

9.3 A finite Wick formula for all the required coefficients

Let $\mathcal{P}(L)$ be the set of set partitions of $\{0, \dots, L-1\}$, and let $(m)_k = m(m-1)\cdots(m-k+1)$. For a partition π , attach independent standard Gaussian vectors to its blocks. Write

$$\Phi_r(\pi) = \mathbb{E} \operatorname{tr} \prod_{t=0}^{L-1} g_{\pi(t)} g_{\pi(t)}^\top.$$

The sample partitions are the classical moment bookkeeping underlying the moment-cumulant calculus [7, Chapter 2]. Wick–Isserlis contraction [3] gives an exact integer polynomial Φ_r . More explicitly, the two occurrences at position t have coordinate indices t and $t+1 \pmod{L}$. Pair Gaussian occurrences only within each block of π . Each pairing equates its two coordinate indices. If the resulting equivalence relation has c coordinate classes, its contribution is r^c . Summing these contributions defines $\Phi_r(\pi)$ without any analytic approximation.

Writing $T_l = \mathbb{E} \operatorname{tr}(\Gamma^l D)$, expansion of the original sample labels yields

$$T_l = \frac{1}{m^{l+2}} \sum_{\pi \in \mathcal{P}(l+1)} (m)_{|\pi|} \Phi_r(\pi) \prod_{B \in \pi} \mu_{|B|+1_{\{0 \in B\}}}, \quad M_j = \sum_{l=0}^j (-1)^{j-l} \binom{j}{l} T_l. \quad (9.4)$$

The extra unit in the moment order of the marked block comes from w_0^2 rather than w_0 . It is exactly the dependence that an independent-copy replacement loses.

For $r = 2$, group the terms in (9.4) by the number k of distinct sample labels. The coefficient of $(m)_k / m^{l+2}$ in T_l is as follows.

$l \backslash k$	1	2	3	4	5
0	6				
1	120	6			
2	5040	312	6		
3	362880	21240	576	6	
4	39916800	2041200	57096	912	6

This is a finite Wick evaluation on at most five sample positions. For example, the one-label coefficient is $\mu_{l+2} \mathbb{E} \|g\|^{2l+2} = (2l+3)!$. For $l = 2, k = 2$, the three partitions contribute $120 +$

$120 + 72 = 312$. The stated coordinate-equivalence rule computes every other entry using only pairings and integer arithmetic. Substituting this table into (9.4) gives the five mixed identities. The accompanying script implements exactly this finite rule and verifies independent small-instance Gaussian-quadrature checks.

9.4 The fourth Gram moment

For independent centered matrix copies, the four-position label expansion gives

$$\mathbb{E} \operatorname{tr} \Delta^4 = \frac{a_r + (m-1)b_r}{m^3},$$

where $a_r = \mathbb{E} \operatorname{tr} X^4$, $b_r = 2 \operatorname{tr} V^2 + \mathbb{E} \operatorname{tr}(XX'XX')$, and $V = \mathbb{E} X^2$. In the current model,

$$\begin{aligned} V &= (3r+5)I, \\ a_r &= r\{105(r+2)(r+4)(r+6) - 60(r+2)(r+4) + 18(r+2) - 3\}, \\ b_r &= r(18r^2 + 81r + 93). \end{aligned}$$

To check the alternating term without commuting matrices, for every symmetric H ,

$$\mathbb{E}[XHX] = 5H + 3(\operatorname{tr} H)I.$$

It follows that the alternating contribution is $5r(3r+5) + 3[3r(r+2) - r^2]$. For $r=2$, $a_r = 37578$ and $b_r = 654$, proving (9.1). This is the *same* ordered independent-sample identity used in Section 4, now with different one-sample moments. The component-reuse interface in [12] organizes this classical sample expansion; neither the partition counting nor Wick pairing is new.

10 The Wick–Hermite interface for the mixed chain

Use the original Gaussian probability space of all (h_i, g_i, ξ_i) , and let $\mathcal{H} = \mathbb{R}^r \otimes L^2$. Let $\mathcal{M}_\Delta$ be matrix multiplication by Δ , and let $c = \mathcal{M}_c \Omega$ be the actual vector-polynomial seed. Then

$$M_j = \langle c, \mathcal{M}_\Delta^j c \rangle_{\mathcal{H}}.$$

In a normalized Hermite basis, scalar coordinate multiplication obeys

$$xH_k = \sqrt{k+1}H_{k+1} + \sqrt{k}H_{k-1}.$$

The seed has total degree at most four and involves at most one sample per term. Multiplication by Δ raises degree by at most four and activates at most one further sample. Let Π retain total degree at most twelve and sample support at most three; one may additionally retain the exact odd residual-coordinate sector and even h_i -parities. With $K = \Pi \mathcal{M}_\Delta \Pi$, both $Kc = \mathcal{M}_{Ec}$ and $K^2c = \mathcal{M}_\Delta^2 c$ hold exactly. Since K is self-adjoint,

$$M_j = \langle c, K^j c \rangle \quad (0 \leq j \leq 4), \quad M_2 = \|Kc\|^2, \quad M_4 = \|K^2c\|^2. \quad (10.1)$$

The odd moments are the corresponding cross inner products. The degree-twelve cutoff suffices because only two forward steps are needed to form the fourth seeded moment; its polynomial expectation contains primitive moments up to degree twenty-four. This is the finite Hermite compression principle in [12], with a non-vacuum seed. The Hermite recurrence and Gaussian chaos interpretation are classical [8]. Exact finite compression is used here, not a bound on the full Gaussian multiplication operator.

The ξ_i -label is preserved under every Δ -insertion because Δ contains no ξ -variables. Its endpoint contraction is (9.2). Wick's rule then evaluates the retained coordinate Grams and sample-label

partitions in (9.4). Finally, the polynomial functional calculus in the next section converts these mixed energies to a reconstruction bound.

No global multiplication norm is estimated. No positive-coefficient cone is assumed: the inverse polynomial has signed coefficients. Wick and the Hermite calculus are compatible exact representations; the framework is not claimed to be mathematically necessary for these finite calculations. Its concrete role here is to retain the seed, its live sample label, and the signed mixed terms until the final valid spectral bound.

11 Signed finite remainder and good-event certificates

Let $\mathcal{A}_\theta = \{\|\Delta\| \leq \theta\}$, where $0 < \theta < 1$, and define

$$p_3(x) = 1 - 2x + 3x^2 - 4x^3, \quad b_\theta = \frac{5 - 4\theta}{(1 - \theta)^2}, \quad q_\theta(x) = p_3(x) + b_\theta x^4.$$

The finite scalar identity and its use on a self-adjoint spectrum are ordinary polynomial algebra and spectral functional calculus; see [17, Chapter 1] for the matrix-function background. They are not a new inverse-moment theorem. The exact finite identity is

$$\frac{1}{(1+x)^2} = p_3(x) + x^4 \frac{5+4x}{(1+x)^2}. \quad (11.1)$$

The final quotient is decreasing for $x > -1$, so on $[-\theta, \theta]$,

$$(1+x)^{-2} \leq q_\theta(x).$$

Importantly, q_θ is nonnegative on *the whole real line*, not merely on the good interval. Indeed $b_\theta \geq 5$ and

$$p_4(x) := p_3(x) + 5x^4 = 5 \left(x^2 - \frac{2}{5}x\right)^2 + \frac{11}{5} \left(x - \frac{5}{11}\right)^2 + \frac{6}{11} > 0.$$

Therefore functional calculus gives

$$\mathbb{E}[F\mathbf{1}_{\mathcal{A}_\theta}] \leq M_0 - 2M_1 + 3M_2 - 4M_3 + b_\theta M_4. \quad (11.2)$$

The global positivity is what permits dropping the indicator from the polynomial on the right; no conditioning-independence step is used.

Theorem 11.1 (Explicit rank-two reconstruction certificate). *For $r = 2$, choose $\theta = 1/2$, and define*

$$d_m = \frac{10464}{m^2} + \frac{590784}{m^3},$$

$$u_m = \frac{6}{m} - \frac{30}{m^2} + \frac{16332}{m^3} + \frac{4732416}{m^4} + \frac{455813568}{m^5}.$$

Then $\mathbb{P}(\mathcal{A}_{1/2}^c) \leq d_m$ and $\mathbb{E}[F\mathbf{1}_{\mathcal{A}_{1/2}}] \leq u_m$. If independent reconstruction sketches are drawn until $\mathcal{A}_{1/2}$ holds and $d_m < 1$, then the accepted reconstruction satisfies

$$\mathbb{E}F_{\text{accepted}} \leq \frac{u_m}{1 - d_m}, \quad \mathbb{E}[\text{number of attempts}] \leq \frac{1}{1 - d_m}. \quad (11.3)$$

The corresponding total squared error is at most $1 + u_m/(1 - d_m)$ for unit residual energy.

Proof. Markov's inequality and (9.1) give $\mathbb{P}(\mathcal{A}_{1/2}^c) \leq 16\mathbb{E} \operatorname{tr} \Delta^4 = d_m$. Here $b_{1/2} = 12$, so substitution of Theorem 9.1 into (11.2) gives u_m . The accepted draw has the original distribution conditioned on $\mathcal{A}_{1/2}$, and its waiting time is geometric. Dividing by the acceptance probability proves (11.3). $\square$

A restart is an explicitly declared algorithmic variant. It requires another reconstruction sketch or precomputed independent candidates; it is not free in a strict single-pass model. The good-event inequality itself applies to the original unchanged sketch. The next section treats its unconditional expectation without restart.

11.1 Comparison on the same good event

The separate-norm argument gives only

$$F\mathbf{1}_{\mathcal{A}_{1/2}} \leq 4\|c\|^2\mathbf{1}_{\mathcal{A}_{1/2}}, \quad \mathbb{E}F_{\text{accepted}} \leq \frac{24}{m(1-d_m)}.$$

The following are outward-rounded upper bounds; all underlying comparisons use exact rational arithmetic.

m	Bad-event bound	Mixed accepted penalty	Separate-norm penalty
500	0.046583	0.012693	0.050346
1000	0.011055	0.006059	0.024269
5000	0.000424	0.001200	0.004803
10000	0.000106	0.000600	0.002401

The comparison fixes the same acceptance event and probability accounting. The near-factor-four difference is not an impossibility theorem for every optimized norm-based proof: a different threshold can improve a norm certificate. The more invariant distinction is that norm separation cannot reproduce the signed mixed coefficients.

12 Unconditional expectation: a separate inverse bound

We now justify an expansion for the *original unrestarted* reconstruction. This step does not follow from positive moments alone.

Lemma 12.1 (Uniform inverse eighth moment). *Put*

$$c_* := \mathbb{E}|h|^{-1/2} = 2^{-1/4} \frac{\Gamma(1/4)}{\sqrt{\pi}}, \quad L_r := 2r(r+2)^5 c_*^4.$$

For every integer $m \geq 32(r+2)$,

$$(\mathbb{E}\|\Gamma^{-1}\|^8)^{1/8} \leq L_r. \quad (12.1)$$

The constants are deliberately crude and are not used in the numerical table. The Wishart input is classical; the elementary batching argument below supplies a sufficient bound for this weighted model, not an optimal integrability threshold.

Proof. Let $b = r + 2$, and form a batch $S_b = \sum_{i=1}^b h_i^2 g_i g_i^\top$, with $s_b = \lambda_{\min}(S_b)$. If $W_b = \sum_{i=1}^b g_i g_i^\top$, then

$$S_b \succeq \min_i h_i^2 W_b.$$

Thus, by independence and concavity,

$$\mathbb{E}s_b^{-1/4} \leq bc_*(\mathbb{E} \operatorname{tr} W_b^{-1})^{1/4} = bc_* r^{1/4}.$$

The last expectation is the standard inverse-Wishart first moment [15, Proposition 10.2] (also [13, Fact 5.3]). For completeness, its value is r : the reciprocal of a diagonal entry of W_b^{-1} is a Gaussian-column residual squared norm with $b - r + 1 = 3$ degrees of freedom, whose reciprocal expectation is one. This follows by the Schur complement and the elementary chi-square integral. Take 32 independent batches. The smallest eigenvalue of their sum is at least the sum of their smallest eigenvalues. Arithmetic–geometric mean gives

$$\mathbb{E} \left\| \left(\frac{1}{32b} \sum_{i=1}^{32b} h_i^2 g_i g_i^\top \right)^{-1} \right\|^8 \leq b^8 (\mathbb{E}s_b^{-1/4})^{32} \leq b^{40} c_*^{32} r^8.$$

For general $m \geq 32b$, split off complete blocks of size $32b$ and drop the nonnegative remainder. These complete blocks retain at least half the mass. Their average has smallest eigenvalue at least the average of their individual smallest eigenvalues; convexity of $x \mapsto x^{-8}$ then supplies at most another factor 2^8 . Taking an eighth root proves (12.1). $\square$

12.1 Explicit positive-moment constants

For even p , let β_p count set partitions of p positions with no singleton blocks, and write

$$\nu_p = (2p-1)!!, \quad R_k(r) = \prod_{a=0}^{k-1} (r+2a).$$

In particular $\beta_8 = 715$, $\beta_{10} = 17722$, and $\beta_{12} = 580317$. Define

$$e_p = \beta_p^{1/p} \left[(\nu_p R_p(r))^{1/p} + \sqrt{r} \right],$$

$$c_p = \left[\beta_p \nu_p R_{p/2}(r) (p-1)!! \right]^{1/p}.$$

Then

$$\|\Delta\|_{L_p} \leq \frac{e_p}{\sqrt{m}}, \quad \|c\|_{L_p} \leq \frac{c_p}{\sqrt{m}}. \quad (12.2)$$

To verify this, for independent identically distributed centered vectors Z_i in any real Hilbert space, expand $\|\sum_i Z_i\|^p$ as a product of inner products. A singleton sample label vanishes. Each surviving term has absolute expectation at most $\mathbb{E}\|Z_1\|^p$, and at most $p/2$ distinct labels. There are at most $\beta_p m^{p/2}$ such assignments. Hence

$$\mathbb{E} \left\| \frac{1}{m} \sum_i Z_i \right\|^p \leq \beta_p m^{-p/2} \mathbb{E}\|Z_1\|^p.$$

Apply this in Frobenius space to $Z_i = X_i$, using $\|X_i\|_F \leq w_i \|g_i\|^2 + \sqrt{r}$, and to $Z_i = w_i g_i \xi_i$. Their local moments are exactly bounded by the expressions defining e_p, c_p .

Theorem 12.2 (Unconditional expansion with finite remainder). *Let $P_4(m, r) = \sum_{j=0}^4 (-1)^j (j+1)M_j$. For $m \geq 32(r+2)$, the original unrestarted penalty is integrable and*

$$|\mathbb{E}F - P_4(m, r)| \leq L_r^2 c_8^2 \left(\frac{6e_{10}^5}{m^{7/2}} + \frac{5e_{12}^6}{m^4} \right). \quad (12.3)$$

In particular, for each fixed r ,

$$\mathbb{E}F = \frac{3r}{m} - \frac{3r(r+3)}{m^2} + \frac{r(45r^2 + 324r + 471)}{m^3} + O_r(m^{-7/2}). \quad (12.4)$$

For $r = 2$, this is

$$\mathbb{E}F = \frac{6}{m} - \frac{30}{m^2} + \frac{2598}{m^3} + O(m^{-7/2}).$$

The remainder is a fixed-rank estimate, not a sharp bound uniform in growing r .

Proof. For every $x > -1$,

$$\frac{1}{(1+x)^2} - p_4(x) = -x^5 \frac{6+5x}{(1+x)^2}.$$

Since Γ is positive definite, this is a globally valid matrix identity on its spectrum. Therefore

$$|F - c^\top p_4(\Delta) c| \leq \|\Gamma^{-1}\|^2 \|c\|^2 (6\|\Delta\|^5 + 5\|\Delta\|^6).$$

Apply Hölder with exponents 4, 4, 2, followed by (12.1) and (12.2). This proves (12.3) and also integrability. The exact polynomial in Appendix B gives (12.4). $\square$

13 Comparison, cancellation, and scope of the mixed result

For rank two, the true mixed correction begins

$$M_0 - 2M_1 + 3M_2 = \frac{6}{m} - \frac{30}{m^2} + \frac{14220}{m^3}.$$

An independent-copy replacement of Δ makes M_1 zero. It leaves the leading term of M_2 equal to $66/m^2$, and therefore gives the false second-order prediction $6/m + 198/m^2$ for the original problem. The lost contribution is $228/m^2$. The true original-estimator expansion in Theorem 12.2 verifies the negative sign without relying on good-event conditioning.

For comparison, the exact fully Gaussian benchmark is

$$\mathbb{E}F_{\text{Gaussian}} = \frac{2}{m-3} = \frac{2}{m} + \frac{6}{m^2} + \frac{18}{m^3} + O(m^{-4})$$

for rank two [16, Lemma A.5]; see also [14, Lemma 1 in arXiv v2] and [13, Theorem 2.1]. The qualitative fact that weighting cannot improve this conditional Gaussian prediction risk also has an established explanation. Let Z_0 have rows $g_i^\top$ and let $H = \text{diag}(h_1, \dots, h_m)$, invertible almost surely. Then

$$\mathbb{E}_\xi F = \|(HZ_0)^\dagger H\|_F^2 \geq \|Z_0^\dagger\|_F^2$$

by the “weighting hurts” inequality [13, Lemma 5.8]. Taking expectations gives $\mathbb{E}F \geq r/(m-r-1)$ for $m > r+1$, with the inequality understood in the extended sense if needed. That comparison does not supply the mixed coefficients in Theorem 12.2. The common-factor geometry has leading inflation three. This does not contradict the empirical universality observations for other tested embeddings in that paper, and it does not decide Conjecture 7.1 (all-algorithm optimality) or the separate generalized-Nyström nonoptimality question in [13, Section 7].

The bounded P02 task is resolved here for a specified structured sketch and subspace-residual family: exact mixed terms, signed finite-remainder control, a nonasymptotic good-event/restart certificate, and a separately justified original-law unconditional expansion. No general inverse-moment calculus, arbitrary-geometry universality theorem, optimal row count, or publication-priority claim is made.

What Chain C adds to the two-proof comparison

For the rMPS scalar extremum, Wick and conditional energy gave the same constants. In this mixed problem their roles are complementary: the Hermite compression keeps an exact seeded moment, Wick evaluates its marked sample partitions, and a nonnegative polynomial majorant preserves the signed combination. Separate norms give a valid but coarser bound on the same event. No comparison here shows that Wick alone cannot evaluate these moments, or that the framework alone supplies inverse integrability.

A Certificate checks and reproducibility

A.1 Exact arithmetic for the reported probe counts

For integer n, χ and $p \leq 4$, compute

$$\mu_p = \left[\prod_{j=0}^{p-1} \frac{\chi + \eta j}{\chi} \right]^{n-1}$$

as a rational number, not from a floating-point exponential approximation. Use these exact values to form A, B from (1.9), (1.10), or (6.7). For rational ε, δ , define

$$F(N) = \frac{A + (N-1)B}{N^3}, \quad H = \delta\varepsilon^4.$$

The table is generated by doubling an upper bracket until $F(N) \leq H$, then binary-searching the bracket. Both $F(N) \leq H$ and $F(N-1) > H$ are checked exactly at the returned integer.

For the listed parameter sets, $3A \geq B \geq 0$ is also checked exactly. This justifies monotonicity on $N \geq 1$, since

$$F'(x) = \frac{-2Bx - 3(A-B)}{x^4} \leq 0 \quad (x \geq 1).$$

The second-moment entry is the rational ceiling in (6.10). Thus these entries are minima of the *specified certificates*, not minima of the true probability of embedding failure.

A.2 Independent normalization and original-core tests

The previously supplied `rmips_p10_checks.py` implements these rational calculations and tensor Gauss–Hermite consistency tests for small original-core instances. It includes real and complex cores, lengths two and three, bond dimension two, and a non-product real two-dimensional coefficient subspace. The isotropy, one-probe fourth coefficient, and alternating-pair coefficient are checked separately.

The degree- $2k-1$ exactness of k -node Gauss–Hermite quadrature is classical; see Golub–Welsch [11] for its Jacobi-matrix construction. Five nodes integrate a real standard Gaussian polynomial through degree nine, whereas these test integrands have degree at most eight in each primitive coordinate. The implementation evaluates quadrature in floating point, so those tests are numerical consistency checks, not exact-arithmetic proof certificates. The rational probe-count checks are exact. The analytic proofs above do not depend on either finite test family.

Additional transfer checks enumerate the 105 real pairings and 24 complex permutations at eighth order. They give the component-count vectors $(1, 12, 44, 48)$ and $(1, 6, 11, 6)$, respectively, agreeing with (2.5)–(2.6). The Gaussian-limit constants $(A_{\mathbb{R}}, B_{\mathbb{R}}) = (234, 46)$ and $(A_{\mathbb{C}}, B_{\mathbb{C}}) = (54, 18)$ are verified by direct substitution. These finite checks validate normalization and implementation, not all parameter values of the theorem.

Checks repeated for this integration. The original rMPS checker is not part of the supplied source bundle, so its reported original-core quadrature experiments have not been rerun in this integration. The included integration checker independently verifies the rational rMPS certificates and local pairing row sums. It additionally runs the supplied P02 checker and a separate seeded Hermite calculation; see Appendix C.

B Mixed reconstruction: exact general-rank polynomials

For the same common-factor family with any $r \geq 1$, equation (9.4) gives

$$\begin{aligned}
 M_0 &= \frac{3r}{m}, \\
 M_1 &= \frac{r(15r + 27)}{m^2}, \\
 M_2 &= \frac{r(9r + 15)}{m^2} + \frac{r(105r^2 + 591r + 768)}{m^3}, \\
 M_3 &= \frac{r(135r^2 + 666r + 807)}{m^3} \\
 &\quad + \frac{r(945r^3 + 10890r^2 + 39069r + 42120)}{m^4}, \\
 M_4 &= \frac{r(54r^2 + 243r + 279)}{m^3} \\
 &\quad + \frac{r(1710r^3 + 17253r^2 + 56757r + 58104)}{m^4} \\
 &\quad + \frac{r(10395r^4 + 202410r^3 + 1393263r^2 + 3938400r + 3756780)}{m^5}.
 \end{aligned}$$

Also

$$\mathbb{E} \operatorname{tr} \Delta^4 = \frac{r(18r^2 + 81r + 93)}{m^2} + \frac{r(105r^3 + 1182r^2 + 4197r + 4500)}{m^3}.$$

The polynomial used in Theorem 12.2 is

$$\begin{aligned}
 P_4(m, r) &= \frac{3r}{m} - \frac{3r(r + 3)}{m^2} + \frac{r(45r^2 + 324r + 471)}{m^3} \\
 &\quad + \frac{r(4770r^3 + 42705r^2 + 127509r + 122040)}{m^4} \\
 &\quad + \frac{5r(10395r^4 + 202410r^3 + 1393263r^2 + 3938400r + 3756780)}{m^5}.
 \end{aligned}$$

For $r = 2$,

$$P_4(m, 2) = \frac{6}{m} - \frac{30}{m^2} + \frac{2598}{m^3} + \frac{1172076}{m^4} + \frac{189922320}{m^5}.$$

C Mixed reconstruction: reproducibility and check limits

The original accompanying `p02_mixed_reconstruction_checks.py` is included unchanged in the reproducibility bundle. It:

1. enumerates the exact sample partitions and Wick pairings in (9.4);
2. checks the general-rank single-sample formulas against direct chi-square moments;
3. independently integrates rank-two, one- and two-sample polynomial moments with seven-node Gauss–Hermite quadrature in each remaining Gaussian coordinate, after exact integration of the residual variables;
4. verifies the scalar remainder and sum-of-squares identities symbolically;
5. computes the restart and separate-norm certificates as exact rational numbers.

The node count is sufficient because the largest power in an individual h_i is twelve and the remaining coordinate powers are no larger; seven Gauss–Hermite nodes integrate through degree thirteen. This is the classical quadrature exactness property [11]. The evaluation is numerical, not a claim of exact rational quadrature output. The quadrature rule has sufficient polynomial degree for these checks; floating-point evaluation produced relative discrepancies below 2×10^{-12} . It is not a simulation. These checks do not by themselves verify the analytic inverse-tail argument or establish novelty; the proofs above supply the analytic claims.

The integration companion `integrated_proof_checks.py` adds an independent *deterministic Hermite-multiplication* calculation for $r = 2$ and $m = 1, 2, 3$. It uses probabilists’ unnormalized Hermite polynomials,

$$x \text{He}_k = \text{He}_{k+1} + k \text{He}_{k-1}, \quad \langle \text{He}_\alpha, \text{He}_\beta \rangle = \mathbf{1}_{\alpha=\beta} \prod_t \alpha_t!.$$

All coefficients and inner products are integers before division by powers of m . It constructs c , Δc , and $\Delta^2 c$ using the original (h_i, g_i, ξ_i) registers, checks the degree and active-sample budgets, and recovers $M_0, \dots, M_4$ as seeded inner products. This independently checks the numerical coefficients in the Wick calculation against the multiplication representation. It is a finite-instance check, not a formal proof for arbitrary parameters. The companion also rechecks the original rMPS rational probe-count tables and pairing row sums. Both scripts state their requirements and write machine-readable JSON reports.

D Attribution map and boundaries of the integrated result

The integration preserves both original rMPS proof chains and adds the P02 calculation as a *different endpoint*. Classical ingredients are cited at their points of use. The following map separates antecedents, calculations proved here, and statements not being made.

Ingredient	Attribution	Boundary
Gaussian tensor-train projections and fourth-order energy iteration	Rakhshan–Rabusseau [2, Theorem 1, Section 5.1].	The one-core strategy and sketch family are not introduced here.
Exact rMPS fourth-moment transfer and surrogate	Camaño–Epperly–Meyer–Tropp [1, Section 4].	Their stated surrogate is fourth-order; higher-order formulas here are derived, not attributed to a published eighth-order theorem.
Gaussian pairing and chaos	Isserlis [3], Goodman [4], Janson [8].	Wick and Hermite are compatible classical descriptions of the same Gaussian law.
Pairing Gram kernels; quadratic-form cumulants	Collins–Matsumoto [5]; Lancaster and McCullagh [6, 7].	No new Haar-integration, pairing, or cumulant theorem is claimed.
Original-law finite compression and retained registers	Heidary’s supplied v11 manuscript [12].	An organizing interface, not sole ownership of conditioning, Wick pairing, or spectral calculus.
Fixed-range reconstruction; current problem context	Tropp et al. [16, Section 4, Lemmas A.3–A.5]; Epperly–Webber [13, Sections 5 and 7].	The algorithm, decomposition, and qualitative weighting loss predate this calculation.
Real iid Gaussian reconstruction benchmark	Tropp et al. [16, Lemma A.5]; Bartan–Pilanci [14, Lemma 1, arXiv v2]; Gaussian inputs in [15].	The reconstruction identity predates the regression restatement and the 2026 benchmark.
Finite inverse polynomial and spectral majorant	Elementary algebra; standard functional calculus, e.g. Higham [17].	The explicit majorant is verified here. No inverse integrability follows from polynomial moments alone.
P02 marked moments, constants, and weighted inverse estimate	Derived in Sections 9–12.	Analytic, self-checked calculations for the stated geometry; not an exhaustive priority claim or an optimized inverse threshold.
Numerical consistency and exact arithmetic	Classical quadrature construction [11]; bundled scripts.	Exact symbolic/rational checks are separated from floating-point quadrature; neither is a Lean certificate.

Version and scope notes. Bartan–Pilanci’s accessible arXiv v2 numbers the benchmark Lemma 1, not Lemma 2.1 as cited in [13]. The supplied v11 spells its author’s name “Diar Heidary”; that spelling is retained. This source-and-role audit is not an exhaustive priority search or a proof-assistant certificate.

References

- [1] C. Camaño, E. N. Epperly, R. A. Meyer, and J. A. Tropp, *Linear algebra at exponential scale via tensor network dimension reduction*, 2026. [arXiv:2606.15350v1](#). *Role*: the model normalization in Definition 1.1; the fourth-order transfer in Section 4.4.1, equation (4.8), and Theorem 4.3; the fourth-moment fractured-Gaussian surrogate in Section 4.4.2, Theorem 4.4; the complex version in Section 4.6. The higher-order calculation here is an extension, not a theorem attributed to that paper.
- [2] B. T. Rakhshan and G. Rabusseau, *Tensorized Random Projections*, Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics, Proceedings of Machine Learning Research **108** (2020), 3306–3316. [Publisher record and paper](#). *Role*: Gaussian tensor-train projection precedent; Theorem 1 and Section 5.1, especially Lemmas 3–4, give the direct fourth-order predecessor of our one-core contraction and independent-core iteration. The all-order extension is proved in this note.
- [3] L. Isserlis, *On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables*, Biometrika **12**(1–2) (1918), 134–139. [doi:10.1093/biomet/12.1-2.134](#). *Role*: the classical probabilistic Gaussian pairing identity, often called Wick’s rule. It is the local Gaussian input to Chain A and the marked-sample moment evaluation in Chain C.
- [4] N. R. Goodman, *Statistical analysis based on a certain multivariate complex Gaussian distribution (an introduction)*, The Annals of Mathematical Statistics **34**(1) (1963), 152–177. [doi:10.1214/aoms/1177704250](#). *Role*: the classical complex Gaussian/Wishart setting, including the circular covariance structure. Balanced pairings here also follow directly from Isserlis applied to real and imaginary parts.
- [5] B. Collins and S. Matsumoto, *On some properties of orthogonal Weingarten functions*, Journal of Mathematical Physics **50**(11) (2009), 113516. [doi:10.1063/1.3251304](#); [arXiv:0903.5143](#). *Role*: Section 2 records the orthogonal pairing Gram kernel d^{loop} and its unitary balanced-pairing counterpart. Our transfer matrices are normalized versions of these classical Grams; no Weingarten pseudoinverse or Haar integration is used in this proof.
- [6] H. O. Lancaster, *Traces and cumulants of quadratic forms in normal variables*, Journal of the Royal Statistical Society, Series B (Methodological) **16**(2) (1954), 247–254. [doi:10.1111/j.2517-6161.1954.tb00167.x](#). *Role*: the trace-power/cumulant description of real Gaussian quadratic forms, used to explain the equality of the constants. The determinant formula and the complex specialization are derived explicitly in this note.
- [7] P. McCullagh, *Tensor Methods in Statistics*, Chapman and Hall, London, 1987. [Author-hosted first-edition chapters](#). *Role*: Chapter 2, the elementary moment–cumulant calculus underlying independent-label assembly. Our ordered-matrix identity is proved entrywise and is not claimed as a new general cumulant result.
- [8] S. Janson, *Gaussian Hilbert Spaces*, Cambridge Tracts in Mathematics, vol. 129, Cambridge University Press, 1997. [doi:10.1017/CBO9780511526169](#). *Role*: classical Gaussian chaos, Wick/tensor structure, and hypercontractivity; see Chapters 2–5. This is background for the Hermite grading, not attribution of a new cutoff principle.

- [9] E. Nelson, *The free Markoff field*, Journal of Functional Analysis **12**(2) (1973), 211–227. doi:10.1016/0022-1236(73)90025-6. *Role*: classical Gaussian hypercontractivity, used only for the bond-blind total-degree comparison, not for the main rMPS contraction.
- [10] P. Da Pelo, A. Lanconelli, and A. I. Stan, *Hölder–Young–Lieb inequalities for norms of Gaussian Wick products*, Infinite Dimensional Analysis, Quantum Probability and Related Topics **14**(3) (2011), 375–407. doi:10.1142/S0219025711004456; arXiv:1101.2947. *Role*: a later proof of Nelson hypercontractivity; equation (1.8) in the arXiv version. The published title is plural; the preprint is titled *A Hölder–Young–Lieb inequality for norms of Gaussian Wick products*. Neither version is credited with originating hypercontractivity.
- [11] G. H. Golub and J. H. Welsch, *Calculation of Gauss quadrature rules*, Mathematics of Computation **23**(106) (1969), 221–230. doi:10.1090/S0025-5718-69-99647-1. *Role*: the classical Jacobi-matrix construction of Gaussian quadrature rules, relevant only to the numerical consistency checks.
- [12] D. Heidary, *Graded Multiplication Operators for Random-Matrix Moments*, Version 11: exposition and attribution revision, 16 September 2026. Supplied unpublished manuscript `v11_attribution_revised.tex`. *Role*: exact finite moment compression, retained sample registers, and the original-law quadratic-energy interface with component reuse. The author and title are transcribed from the supplied manuscript. These organizing interfaces do not replace the public Gaussian contraction, pairing, or cumulant precedents above.
- [13] E. N. Epperly and R. J. Webber, *Sharp analysis of sketched least squares and randomized low-rank approximation*, 2026. arXiv:2605.19096v2. *Role*: Theorem 2.1 for sharp Gaussian/orthonormal benchmarks; Section 5.1 for the reconstruction error decomposition; Section 5.2 for classical Wishart inputs; Section 7 for the distinct universality and all-algorithm optimality questions. The earlier real regression formula is credited there to Bartan–Pilanci. The matching fixed-range reconstruction identity already appears in Tropp et al. (2017), also cited directly here. The present fixed-geometry calculation does not resolve the Section 7 conjectures.
- [14] B. Bartan and M. Pilanci, *Distributed Sketching Methods for Privacy Preserving Regression*, 2020. arXiv:2002.06538v2. *Role*: Lemma 1 in this version gives the exact real Gaussian sketch-and-solve prediction-error factor $r/(m - r - 1)$; its proof uses the inverse-Wishart expectation. This is a directly checked regression antecedent of the 2026 benchmark, not the earliest fixed-range reconstruction formula and not a claim about the weighted Khatri–Rao calculation.
- [15] N. Halko, P. G. Martinsson, and J. A. Tropp, *Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions*, SIAM Review **53**(2) (2011), 217–288. doi:10.1137/090771806; arXiv:0909.4061v2. *Role*: Propositions 10.1–10.2 give the standard Gaussian conditional Frobenius and pseudoinverse moment ingredients. The inverse-Wishart first moment is classical multivariate statistics, not introduced by this note or by this survey. The weighted inverse-eighth-moment batching bound is proved here.

- [16] J. A. Tropp, A. Yurtsever, M. Udell, and V. Cevher, *Practical sketching algorithms for low-rank matrix approximation*, SIAM Journal on Matrix Analysis and Applications **38**(4) (2017), 1454–1485. doi:10.1137/17M1111590; arXiv:1609.00048v2. *Role*: equations (4.2)–(4.3) for $Q(\Psi Q)^\dagger \Psi A$; Lemmas A.3–A.4 for the error decomposition; Lemma A.5 for the exact conditional Gaussian reconstruction error. Remark 4.1 credits earlier algebraically equivalent algorithms. This is a direct predecessor of the P02 setup and its Gaussian calibration, not of the weighted mixed-moment coefficients.
- [17] N. J. Higham, *Functions of Matrices: Theory and Computation*, Society for Industrial and Applied Mathematics, Philadelphia, 2008. doi:10.1137/1.9780898717778. *Role*: Chapter 1 for standard spectral/polynomial matrix-function calculus. The scalar finite-remainder and sum-of-squares identities used in Chain C are explicitly verified in this note; no new general matrix-function or inverse-series theorem is asserted.